Congestion Spillovers in Multi-Stage Public Services: Evidence from England's Emergency Care System*

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Abstract

In multi-stage public service systems, congestion at one stage can propagate to others, but credible evidence on the magnitude of such spillovers is scarce. We study the emergency healthcare system in England, where hospital discharge delays create upstream congestion that traps ambulances at hospital doors and slows emergency response in the community. We exploit a £700 million discharge fund introduced during the winter of 2022–2023 that targeted hospital patient flow without directly intervening in ambulance services. Using regression discontinuity in time and difference-in-discontinuities design around the structural break in discharge rates, we find that the policy increased return of spontaneous circulation for cardiac arrest patients by 3.8 percentage points and 30-day survival by 2.4 percentage points. The mechanism operated through accelerated patient throughput which reduced ambulance handover delays and freed vehicles for emergency dispatch. With a back-of-the-envelope calculation, we show that the survival gains for cardiac arrest patients alone offset 11% of the programme's cost, given that the cardiac arrest patients represent just 0.6% of all hospital admissions. Our findings shed light on the importance of accounting for cross-stage spillovers when evaluating public service interventions, and suggest that targeted investments to relieve bottlenecks can yield large system-wide benefits in emergency care.

JEL codes: I18, J18; H54.
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1 Introduction

Public services are often delivered through a sequence of connected stages. In some systems, users do not receive the service at a single point. Instead, they move through several linked steps before the service is completed. Healthcare operates in a such structure. A patient may move through emergency call handling, ambulance transport, hospital treatment, and discharge into community or social care. Each stage depends on its own staff, facilities, and equipment, but no stage operates in isolation. The capacity of one part of the system depends on whether the next part is able to receive patients and move them forward. As a result, pressure in one stage can reduce performance in other parts of the system as well.

These systems often operate under persistent congestion. In publicly provided services, resources are limited relative to demand, and access is often rationed through waiting rather than prices (Gravelle and Siciliani, 2007). The direct effects of congestion at a single stage are familiar: longer delays, slower treatment, and lower service quality. But in a system made up of connected stages, congestion is not only a local problem. When one part of the system slows down, the effects can spread to other parts. Upstream units may be unable to move users forward, while downstream units may receive users later or in worse condition. For this reason, the cost of congestion at one stage may be greater than it first appears. Looking only at the disrupted stage may miss wider losses elsewhere in the system. By the same logic, the benefits of relieving congestion at one stage may also extend beyond that stage alone.

This has an important implication for public resource allocation. When an intervention eases congestion in one part of a connected system, it may improve performance in other parts as well. The return to such an investment may therefore be larger than a stage-by-stage evaluation would suggest. At the same time, these wider gains are unlikely to be the same at every point in the system. Relieving congestion in one stage may generate only modest benefits elsewhere, while relieving it in another may improve performance across the system more broadly. Whether such cross-stage spillovers exist, and how large they are, is therefore central to evaluating any capacity investment in a congested public service system, while yet credible evidence remains scarce, because congestion at different stages is jointly determined and policy rarely shifts capacity at one stage in isolation.

This paper provides quasi-experimental evidence on the existence of cross-stage congestion spillovers in a major public service system. We study the emergency healthcare system in England, where hospital bed shortages, which are driven by the accumulation of patients who no longer meet the criteria to reside in the hospital but cannot be safely discharged, create a chain of

upstream congestion that ultimately degrades the quality of pre-hospital emergency care (Mason *et al.*, 2016; Howlett *et al.*, 2026). When beds are occupied by patients awaiting discharge, emergency admissions cannot flow through, ambulances queue at hospital doors waiting to hand over patients, and fewer ambulances are available to respond to life-threatening emergencies in the community. The final link in this chain concerns patients waiting for emergency response in the community. Delays may matter for many ambulance patients, but they are especially consequential for time-critical conditions such as out-of-hospital cardiac arrest, where survival falls sharply with each minute of delay.

We exploit a large-scale policy intervention that targeted the downstream end of this chain (Department of Health and Social Care, 2023). During the winter of 2022–2023, the UK government allocated approximately £700 million in emergency discharge funding. This funding comprised an initial £500 million Adult Social Care Discharge Fund and an additional £200 million announced on 9 January 2023 in order to accelerate the movement of medically fit patients out of hospitals and into community-based step-down care. The fund was centrally allocated as additional emergency spending, rather than drawn from existing local health budgets, which mitigates the concern that spending levels reflect unobserved local characteristics or discretionary priority-setting by local decision-makers. Crucially, the intervention was designed to relieve congestion at the discharge stage; it did not directly target ambulance services, emergency departments, or pre-hospital care. This feature is what makes the setting valuable for identifying cross-stage spillovers: any improvement observed in pre-hospital emergency outcomes cannot be attributed to direct investment in that stage, and is consistent with propagation through the congestion channel, provided no other factor changed discontinuously.

Our identification strategy exploits the sharp timing of the discharge improvement. Using daily hospital discharge data, we document that the rate at which medically fit patients were discharged underwent a structural break in the first week of January 2023, recovering from its winter trough and remaining elevated thereafter. We implement a regression discontinuity in time (RDiT) design around this cutoff, comparing out-of-hospital cardiac arrest outcomes in narrow windows before and after the point at which hospital discharge capacity systematically improved. Because the funding was deployed during an acute system crisis, patients could not anticipate or time their need for emergency care around the improvement as cardiac arrests are by nature unpredictable, which supports the validity of the RDiT design. To rule out that the effect of the discharge improvement simply reflects seasonal or institutional shocks that coincided with the policy, we implemented a difference-in-discontinuities (DiDC) framework, which differences out any discontinuity observed at the same calendar date in a comparison year unaffected by the pol-

icy.

We find that the discharge improvement produced significant gains in cardiac arrest outcomes upstream of the intervention point. The return of spontaneous circulation (ROSC), a clinical marker of whether a patient's heart is successfully restarted, increased by 3.8 percentage points in our preferred DiDC specification, and 30-day survival improved by 2.4 percentage points. These effects are robust to alternative bandwidths, donut specifications that exclude observations immediately adjacent to the cutoff, and placebo cutoffs at the same calendar date in 2022 and 2024. We trace the mechanism through two links in the congestion chain. First, the policy produced a discontinuous increase in hospital discharge rates, indicating a structural improvement in patient flow. Second, this was followed by a sharp reduction in ambulance handover delays which is the operational bottleneck through which discharge-stage congestion propagates upstream into emergency response times. We did not find a structural break in the number of influenza beds, which implies the above effects are not owing to normal pandemic of infectious diseases. Notably, we find no corresponding discontinuity in hospital bed availability, suggesting that the policy operated by accelerating patient throughput rather than by expanding hospital bed capacity. These patterns suggest that the relevant constraint was not simply the stock of beds, but the rate at which they could be turned over.

The magnitude of these cross-stage spillovers is economically meaningful, even when measured through a single narrow channel. In our data, cardiac arrest patients constitute only 0.6% of total hospital admissions from January to March 2023, yet a back-of-the-envelope cost-effectiveness calculation within that time period shows that the survival gains in this subgroup alone account for approximately 11% of the programme's £700 million cost. If the gains accruing to the remaining 99.4% of emergency patients were even modestly positive, the aggregate return would be substantially larger though we do not estimate those additional benefits directly and so treat the cardiac arrest channel as a conservative lower bound. This calculation illustrates a core empirical point of the paper: the cardiac arrest patients in our sample benefited from the discharge policy not through their own hospital discharge, but through an entirely upstream channel that faster ambulance response made possible by the decongestion of hospital handover bays. A conventional evaluation that assessed the discharge fund in terms of hospital throughput, length of stay, or readmission rates would never capture a survival gain that occurred at home or roadside, before the patient ever reached hospital. In interconnected service systems, stage-specific programme evaluations may substantially understate the social returns to capacity investments, precisely because the benefits that propagate to other stages are invisible to an evaluator who looks only locally.

This paper contributes to three strands of literatures. First, we contribute to the economics

of congestion in public services. A growing body of work documents the costs of congestion in individual service settings. This literature originated in transportation, where classic models emphasize that when capacity is fixed, additional users impose delay and scheduling costs on others, generating congestion externalities (Vickrey, 1969; Arnott *et al.*, 1990, 1993). More recently, economists have examined congestion in public health systems, focusing on at least three related margins: rationing by waiting in publicly financed care (Gravelle and Siciliani, 2007, 2008a,b), hospital and emergency department crowding and its effects on treatment and mortality (Hamilton and Bramley-Harker, 2001; Hoe, 2022; Gutierrez and Rubli, 2021), and provider responses to binding capacity constraints such as reduced time per patient or altered care intensity (Harris *et al.*, 2020). Related work has also studied congestion in other individualized service settings, including courts (Bielen *et al.*, 2018; Müller, 2022), public administration (Deshpande and Li, 2019; Kearney *et al.*, 2021), and education(Lazear, 2001; McMullen and Rouse, 2012; Phan *et al.*, 2024), where backlogs, wait times, or crowding similarly reduce access, distort behaviour, or lower service quality.

While these studies typically examine congestion within a single stage, our paper provides evidence that the welfare costs of congestion extend beyond the stage where it binds, through a mechanism of backward propagation in which downstream blockages deteriorate upstream performance. The theoretical foundations for this mechanism are well established in operations research, where models of tandem queuing networks with finite buffers show that when a downstream station is saturated, upstream stations are "blocked", which means they are forced to hold units that cannot advance, therefore reducing their own effective capacity (Perros, 1994; Balsamo *et al.*, 2001). Applied to healthcare, Koizumi *et al.* (2005) demonstrate through simulation that downstream capacity shortages in a patient flow network generate upstream blocking that congests the entire system, and that relieving the specific bottleneck is the most cost-efficient intervention that is a prediction our quasi-experimental evidence directly confirms. Zychlinski *et al.* (2020) model a closely related problem, showing how scarce capacity in post-acute geriatric institutions creates bed blocking that propagates back into hospital wards. Within the hospital, Kim *et al.* (2024) provide empirical evidence that utilisation shocks in one inpatient unit spill over to neighbouring units, and in concurrent clinical work, Howlett *et al.* (2026) document correlational evidence that medical patient boarding in English emergency departments is associated with longer ambulance handover and response times. Our paper complements this body of work in two respects: we exploit quasi-experimental policy variation to make causal claims about the magnitude of cross-stage spillovers, and we trace the congestion chain beyond operational metrics to patient survival outcomes.

Second, we contribute to the literature on the economics of emergency medical services. Prior extensive research has emphasized how factors such as response time targets (Pell *et al.*, 2001), public/private provision (Knutsson and Tyrefors, 2022; Chan *et al.*, 2023), delays in response (Luc-chese, 2024), handover time (Dawson *et al.*, 2022), austerity (Berman and Hovland, 2024), hospital crowding (Hoe, 2022; Jones *et al.*, 2022), incentives in emergency departments (Gruber *et al.*, 2023), and technology adoption (Athey and Stern, 2002) affect performance and patient outcomes. We extend this literature by identifying an upstream determinant of ambulance availability — hospital discharge congestion — that lies outside the emergency care system itself, and by showing that a policy which never directly targeted ambulance services nonetheless produced measurable improvements in pre-hospital cardiac arrest outcomes through the congestion channel. We show that when additional spending is directed toward relieving these bottlenecks, the system can respond more effectively, yielding immediate and measurable improvements in health outcomes. Our findings thus highlight both a novel mechanism and a concrete policy lever linking emergency system performance to population health.

Third, we contribute to the economics of hospital discharge and patient flow. A substantial literature studies the determinants and consequences of delayed discharge in the English NHS and Portuguese healthcare system (Gaughan *et al.*, 2015, 2017; Moura, 2022), and a related strand examines how social care provision interacts with acute hospital performance (Fernandez and Forder, 2008; Crawford *et al.*, 2021; Longo *et al.*, 2023). These studies consistently find that discharge delays and social care shortages worsen hospital-level outcomes including length of stay, readmission rates, bed occupancy, and A&E attendance, but their evaluation frame stops at the hospital boundary. We extend this literature by tracing the effects of discharge-stage relief beyond the hospital and into the pre-hospital emergency system, showing that faster discharge translates into reduced ambulance handover delays, shorter response times, and higher cardiac arrest survival. Moreover, our finding that the discharge fund operated by accelerating patient throughput rather than by expanding bed capacity suggests that, for interconnected care systems under pressure, improving the rate of flow through existing resources may yield larger systemic returns than adding capacity at any single stage.

The remainder of the paper is structured as follows. Section 2 describes the institutional setting of the UK emergency care system and the discharge funding policy. Section 3 presents the data. Section 4 details the empirical strategy. Section 5 reports the results, including the main RDiT and DiDC estimates, mechanism evidence on hospital discharge and handover delays, robustness checks, and cost-effectiveness implications. Section 6 concludes.

2 Background of the English Healthcare System

This section introduces the background needed to understand the UK's urgent and emergency care system and the congestion spillover mechanism. We first describe the patient flow through the system, from the 999 call to hospital discharge, using this structure to introduce where congestion arises and how it propagates. We then describe the recent systemic pressures and shocks, including the COVID-19 pandemic, workforce shortages in social care and nursing, and the 2022–2023 industrial action that exposed and amplified these bottlenecks. Finally, we outline the government's policy response, that is the discharge-targeted funding interventions whose effects we evaluate in the remainder of the paper.

2.1 Patient Flow and the Anatomy of Congestion Spillovers

We organise this subsection by following a patient through the emergency care pathway, from the moment a 999 call is placed to the moment the patient is discharged from hospital. At each stage we introduce the relevant performance measure and identify where congestion can originate. At the end, we draw these pieces together to describe how a downstream hospital discharge bottleneck propagates upstream and consequently affects ambulance response in the community.

(a) Ambulance dispatch and response. Emergency care services in England are organised into regional NHS ambulance trusts, each responsible for emergency medical services across a designated geographical area. There are currently 10 major regional ambulance trusts covering England, with each trust managing its own fleet, staff deployment, and operational protocols while adhering to national performance standards. Since the 2022 Health and Care Act, these trusts work closely with Integrated Care Boards (ICBs), which are responsible for planning and funding health services across their local areas. ICBs hold commissioning responsibility for ambulance services and coordinate between ambulance trusts, hospitals, and other healthcare providers to manage patient flow and resource allocation. Private ambulance services also play a role: when public ambulance resources are fully utilised, NHS trusts can dispatch a contracted private ambulance to respond to a 999 call. Private ambulance crews are expected to adhere to the same clinical standards and protocols, and NHS ambulance trusts document those events in their own data management systems.

Residents access emergency care primarily by dialling 999, though patients can also self-present at hospital Accident and Emergency (A&E) departments. Figure B.2 presents the ambulance dispatch process. Upon receiving a 999 call transferred from the BT operator, call handlers in the local ambulance service's Emergency Operations Centre (EOC) assess the urgency. From the mo-

ment this first call is received, the clock starts on the ambulance response time. The patient or caller answers a series of standardised protocol questions about address, contact numbers, symptoms, medical history, and basic demographic details, providing sufficient information for the call handler to classify the emergency. Calls are divided into four categories according to patient conditions: Category 1 (C1) for life-threatening emergencies (e.g., cardiac arrest), with a target average response time of 7 minutes; Category 2 (C2) for serious but not immediately life-threatening emergencies, with a target average response time of 18 minutes; and Category 3 (C3) and Category 4 (C4) for urgent or less urgent cases requiring assessment or transport, with standards specifying response to 90% of calls within 120 minutes (C3) and 180 minutes (C4), respectively.

This initial triage is a prominent clinical process that determines both the priority of each call and the appropriate response dispatched, including the type of vehicle and the level of paramedic skill required. For C1 and C2 patients with shorter response time standards, resources are dispatched immediately to ensure that the sickest patients are reached the fastest. Once classification is complete, the ambulance crew is mobilised. Upon arrival on scene, the response time clock stops and initial assessment and treatment begin. The crew then transports the patient to hospital.

Ambulance response time is a key measure for the performance of emergency medical services. It directly reflects how quickly potential life-saving medical intervention can begin. For a range of time-sensitive conditions, for instance, the "golden hour" principle in trauma or "time is tissue" in cardiac and stroke care, delays in response are strongly associated with higher patient death risks and/or significant morbidity (Pell *et al.*, 2001; Holmén *et al.*, 2020; Alarhayem *et al.*, 2016). Thus, ambulance response time measures the critical interval from an emergency call to the moment providers reach the patient, which marks the first opportunity to administer formal treatment.

(b) Ambulance-to-hospital handover. Upon arrival at the hospital, the ambulance crew transfers the patient to A&E staff. This handover is a critical transition in the emergency care pathway: NHS England sets a 15-minute target for handover completion, and any handover taking more than 30 minutes is recorded as a "delayed handover". After a successful transfer, the ambulance crew becomes available to respond to the next emergency call, and the patient begins formal hospital care.

When hospitals are congested and operating near full capacity, however, they may be unable to accept incoming patients from ambulances promptly, leading to extended handover delays. Ambulances delayed at hospital become unavailable for subsequent 999 calls, raising response times for patients waiting in the community. We return to the mechanism behind this congestion

below.

(c) Assessment and admission at A&E. After handover, ambulance-arriving patients are assessed in A&E. A&E staff stabilise the patient where necessary and determine the disposition: some patients are treated and discharged directly from A&E, while others require further care and are admitted to an inpatient ward or transferred to a specialist unit.

(d) Inpatient stay and discharge. A patient's hospital stay typically proceeds through admission, assessment, diagnosis, treatment, and discharge. Discharge does not simply mean the patient leaves hospital and returns home without further support. Under the NHS discharge framework, patients are assigned to one of four discharge pathways depending on their post-hospital care needs (Department of Health and Social Care and NHS England, 2024). Pathway 0 refers to patients who require no new or additional support and can return home directly. Pathway 1 includes patients who are discharged home with a package of home-based care to support recovery. Pathway 2 applies to patients who are not yet well enough to return home safely but no longer require acute hospital care; these patients are discharged to a temporary community hospital, rehabilitation unit, or other step-down setting. Pathway 3 is intended for patients with the highest ongoing care needs, typically following a life-changing health event, who require placement in a residential or nursing home. Discharge for more severe cases therefore depends not only on medical readiness but also on the availability of complementary health and social care resources such as home care packages, step-down beds, and long-term care placements.

When these downstream care resources are insufficient, particularly during periods of acute demand pressure, patients who are medically fit for discharge remain in hospital (Gaughan *et al.*, 2015, 2017; Allan *et al.*, 2021; Gridley *et al.*, 2022; Fernandez *et al.*, 2018). This generates delayed transfers of care, often referred to as "bed-blocking," and constitutes a major source of hospital congestion. Such congestion is one of the most serious challenges facing the English healthcare system (Rojas-García *et al.*, 2018). Patients who remain unnecessarily in hospital face higher risks of hospital-acquired infection, mental health deterioration, and physical deconditioning (Chen *et al.*, 2022; Guilcher *et al.*, 2021; Hussein *et al.*, 2024). For hospitals, delayed discharge reduces bed availability, contributes to overcrowding in A&E departments, and imposes substantial avoidable costs, since acute hospital beds are considerably more expensive than community or social care alternatives.

Congestion spillovers across the patient flow. The four stages above are linked by scarce, shared resources, so a bottleneck at any one stage propagates through the others. When social care capacity is insufficient and discharge slows, inpatient wards fill up. With ward beds unavail-

able, A&E's capacity to absorb new arrivals is reduced. Ambulances arriving at the hospital then cannot complete handover promptly, generating extended handover delays (Dantis, 2025; Care Quality Commission, 2021). Ambulances and crews tied up in handover queues are unavailable to respond to subsequent emergencies, raising ambulance response times for patients waiting in the community. For time-critical conditions such as cardiac arrest or stroke, these delays directly worsen patient outcomes (Dawson *et al.*, 2022). Cross-stage congestion spillovers of this form arise naturally when a multi-stage service system shares scarce resources across stages: a hospital discharge bottleneck can generate measurable effects far upstream, at the point of emergency dispatch. We formalise this mechanism in a stylised multi-stage congestion model presented in Appendix A, which derives the propagation of demand shocks through the system and shows that flow-improving interventions (raising the discharge rate) yield dual welfare benefits relative to capacity expansion alone.

2.2 Recent Systemic Pressures and Shocks

The propagation mechanism described above is most consequential when the system is already operating near capacity. Several recent shocks brought the English healthcare system to and beyond this margin.

The COVID-19 pandemic. The COVID-19 pandemic placed unprecedented strain on the UK's emergency medical services from 2020 onward. The crisis manifested through multiple channels: increased demand for urgent care due to COVID-related complications, reduced ambulance capacity as staff fell ill or were required to isolate, and prolonged hospital handover delays as acute-care facilities became saturated. Appendix Figure B.4 illustrates these pressures. The top panel shows a sharp and persistent increase in average response times, with further deterioration after the relaxation of lockdown restrictions in 2021. The middle panel shows rising incident volumes, especially for C1 patients, while the bottom panel reveals elevated mortality rates during peak pandemic periods. In the short term, COVID-19 substantially increased demand for emergency care, leading to hospital congestion and longer discharge delays, as many patients required extended inpatient treatment before meeting discharge criteria. In the longer term, survivors of COVID-19 have faced a higher prevalence of chronic health complications (e.g., respiratory, cardiovascular, and mental health conditions), increasing ongoing demand for emergency and urgent care services (Tsampasian *et al.*, 2024; Adu-Amankwaah, 2024; Daines *et al.*, 2022).

Workforce shortages in social care and nursing. The COVID shock landed on a healthcare and social care system already weakened by chronic workforce pressures. The adult social care workforce in England has faced two persistent problems: high turnover and elevated vacancy

rates. Hoddinott (2025) reports that turnover among social care workers has remained above 29% every year since 2017–2018, peaking at 31.4% in 2019–2020 and remaining at 29.0% in 2022–2023, which is well above the pre-2015 level of around 22–24%. Vacancy rates rose sharply during the pandemic, from 6.8% in 2020–2021 to 10.5% in 2021–2022 and 9.8% in 2022–2023, before falling to 8.3% in 2023–2024 and 7.0% in 2024–2025. These two indicators capture distinct margins of workforce fragility. Sustained high turnover erodes the firm-specific experience and continuity of care that providers can deliver: as experienced workers leave the sector, average tenure falls, providers carry recurrent retraining costs, and patients face frequent changes in carers. Elevated vacancy rates compound this by limiting the absolute capacity available to provide care. Both margins matter for hospital discharge directly: Pathway 1–3 transitions depend on home care packages, step-down beds, and residential placements that an understaffed sector cannot supply at scale, leaving patients in hospital even when medically ready to leave.

Industrial action in 2022–2023. The sustained pressure from the pandemic culminated in the largest ambulance strike in over 30 years during late 2022 and early 2023. Ambulance workers across England and Wales participated in coordinated industrial action involving all three major unions (GMB, UNISON, and Unite). The strikes began on December 21, 2022, and intensified through January 2023, with the largest coordinated action occurring on January 23, 2023. The action was triggered by the government's pay offer, which unions argued represented a real-terms pay cut given high inflation, but it reflected deeper systemic issues including deteriorating workloads and rising vacancy rates in ambulance services. The disputes were resolved in March 2023 following government negotiations that delivered improved pay settlements, though this also led to new minimum service level legislation restricting future strike action in emergency services.

It is noteworthy that the East of England Ambulance Service NHS Trust (EEAST) was exempt from the December 2022 strikes. UNISON's initial ballot for ambulance workers at EEAST failed to meet the 50% turnout threshold required for legal strike action in November 2022 (UNISON, 2023). This regional variation in strike participation provides useful contextual variation across trusts and is revisited in our empirical analysis, where we examine local differences in response times and health outcomes during this period.

2.3 Government Response: The 2022–2023 Discharge Fund

In response to the severe winter pressures on the emergency care system and hospitals in 2022–2023, the UK government introduced a series of policy interventions targeted specifically at the hospital discharge bottleneck. Critically, none of these interventions provided direct funding to ambulance services. This targeting is central to our empirical strategy: any observed improve-

ment in ambulance performance over this period must operate through the discharge channel rather than through direct investment in ambulance capacity.

The first and largest intervention was the £500 million Adult Social Care Discharge Fund, announced in September 2022, with detailed allocation guidelines released in mid-November. In early December 2022, the first tranche of the fund (£200 million, 40% of the total) was distributed to local authorities and ICBs. The remaining £300 million was scheduled for release on 31 January 2023, conditional on meeting specific requirements, including submission of spending plans by 16 December 2022 and subsequent fortnightly activity reports documenting delivery of the planned interventions (Department of Health and Social Care, 2023). Spending under this scheme was directed primarily towards workforce support and service capacity expansion: local areas used the funds to purchase additional staff hours and to raise minimum wages to attract and retain workers in the high-turnover social care sector, and authorities increased capacity by block-purchasing care home beds and additional home care packages. These measures aimed to accelerate patient discharge from hospitals, thereby freeing capacity for incoming patients during a period of heightened seasonal demand. Notably,the fund did not attempt to resolve the underlying structural workforce shortages described in Section 2.2, but rather to ease their immediate consequences during the winter peak through short-term capacity expansion.

A key challenge, however, was the short notice and compressed implementation timeline. After receiving detailed allocations in November, local authorities were required to design spending plans and implement interventions within a matter of weeks. Immediate implementation was often infeasible: home care providers may have responded to sudden demand by raising prices, limiting the effective purchasing power of the fund, and the administrative burden of planning, reporting, and compliance diverted attention from ongoing winter plans. As a result, even where funding was formally allocated, it was unlikely to translate rapidly into observable improvements in system performance.

Figure B.3 illustrates the distribution of actual spending under the Adult Social Care Discharge Fund, covering the reporting period from 23 December 2022 to 31 March 2023. Spending was highest at the beginning and end of the period, consistent with the compressed timelines and final reporting deadlines. In total, £483 million was spent to support the health and social care system in managing the winter crisis.

The second major intervention was announced on 9 January 2023, when Health Secretary Steve Barclay unveiled an additional £200 million emergency funding package targeted specifically at further reducing hospital discharge delays. This scheme operated as a reimbursement-

based programme administered through ICBs, which were required to immediately secure bedded step-down capacity in care homes and other community settings. This direct funding mechanism allowed ICBs to respond flexibly to local demand, complementing the earlier £500 million discharge fund. An additional £50 million in capital funding was allocated to create new ambulance hubs and expand hospital discharge lounges, addressing physical infrastructure constraints that contributed to handover delays.

We now turn to evaluating whether and how much this discharge-targeted funding actually relieved congestion across the multi-stage system, and the magnitude of the resulting improvements in patient outcomes.

3 Data

3.1 Data Sources

We describe each data source in turn, following the patient flow from ambulance dispatch through hospital discharge to health outcomes.

Ambulance response times. Local ambulance service performance is measured by the weekly average ambulance response time and the corresponding number of incidents in each local authority. The ambulance response time is defined as the time elapsed from receiving a 999 call to the arrival of an ambulance vehicle at the patient's location. These data come from Freedom of Information requests we made to each ambulance trust in the UK, covering 10 ambulance trusts in England, Scotland, and Wales.[1] Northern Ireland is excluded from the analysis, and data from the South Western Ambulance Trust and South Central Ambulance Trust were not obtainable. We use the data from 8 trusts in England in the following analysis, which operate within a consistent policy context.

The data are provided at postcode district-by-week granularity, aggregating all incident-level information from a postcode district in a given week. To match the granularity of mortality data, which is at the local authority by week level, we merge the data to local authority level: the number of ambulance incidents is summed across all postcode districts in a local authority, and the average ambulance response times are averaged using the number of incidents as weights. The

[1]Those 10 ambulance trusts are: North East Ambulance Service NHS Foundation Trust; Yorkshire Ambulance Service NHS Trust; North West Ambulance Service NHS Trust; West Midlands Ambulance Service University NHS Foundation Trust; East Midlands Ambulance Service NHS Trust; South East Coast Ambulance Service NHS Foundation Trust; London Ambulance Service NHS Trust; East of England Ambulance Service NHS Trust; The Welsh Ambulance Services NHS Trust which covers the whole of Wales; and The Scottish Ambulance Service which covers the whole of Scotland.

resulting balanced panel covers April 2018 to the end of 2023.

Hospital handover delays. We collect hospital handover data from the NHS England Urgent and Emergency Care Daily Situation Reports (SitRep) dataset. This dataset provides daily data by NHS Foundation Trust in England, including the number of patients arriving by ambulance, the number of ambulance handover delays between 30 and 60 minutes, the number of handover delays greater than 60 minutes, and the total time lost to ambulance handover delays. Although this dataset provides fine daily granularity, it is reported only for the winter months (November to April) of each year. In our analysis, we use data from 14 November 2022 to 2 April 2023, which fully covers the period of the policy intervention.

Hospital bed capacity. We use hospital bed capacity data from the same NHS England Urgent and Emergency Care Daily SitRep dataset. The data are reported daily by NHS Foundation Trust and classified into adult General & Acute beds and adult Critical Care beds, with both total available beds and beds occupied reported. We use these data to test an alternative mechanism through which the discharge fund could have operated: rather than improving patient flow, the fund might have expanded inpatient capacity by opening additional beds. If so, we would expect to observe a discontinuity in available beds at the policy cutoff. We test this in Section 5.4.

Within the same SitRep dataset, we also observe the daily number of beds occupied by patients with laboratory-confirmed influenza, separately for General & Acute and Critical Care wards. We use these influenza bed series in our identification check (Section 5.3.5) to assess whether the cutoff is contaminated by a concurrent change in winter influenza pressure, which would independently affect both hospital demand and patient outcomes.

Hospital discharge. We measure hospital discharge performance using the NHS England Urgent and Emergency Care Daily SitRep dataset, which reports the daily number of patients who no longer meet the criteria to reside in hospital and the number of these patients who were discharged on each day, by NHS Foundation Trust. The criteria to reside framework was introduced in the 2022 *Hospital Discharge and Community Support Guidance* (Department of Health and Social Care and NHS England, 2024) and identifies patients who are medically fit for discharge but remain in hospital, which is the population directly targeted by the discharge fund. We construct our main discharge measure as

$$\text{Hospital Discharge Rate}_{i,t} = \frac{\#\text{ patients discharged}_{i,t}}{\#\text{ patients no longer meeting the criteria to reside}_{i,t}},$$

that is, the daily share of medically-fit-for-discharge patients who are actually discharged in trust i on day t. The national trend in the discharge rate is shown in Figure B.1. Because the series

exhibits pronounced day-of-week cyclicality, we work with its 7-day moving average in the analysis (see Section 5.1). As with the handover delay data, the discharge series is reported only during winter months (November to April), and we use the same November 2022 to April 2023 window for analysis as well as the data in the last year. We obtained all outcomes by full group and group of patients who have length of stay more than 21 days.

Cardiac arrest outcomes. To examine clinically relevant health outcomes attributable to changes in the emergency care system, we use the NHS England Ambulance Quality Indicators (AQI) dataset. This dataset reports two key cardiac arrest outcomes monthly at the ambulance trust level from January 2021 to February 2025. The first is the *Return of Spontaneous Circulation* (ROSC) rate, the proportion of cardiac arrest patients whose pulse is restored by the time of hospital arrival. ROSC measures the success rate of the ambulance crew's resuscitation efforts and is therefore an indicator of pre-hospital emergency care system performance: when ambulance response times are shorter, patients receive life-saving interventions sooner, increasing the likelihood of achieving ROSC. The second is the *30-day survival rate* following cardiac arrest, which captures whether patients ultimately survive the event. Achieving ROSC is only the initial step in recovery; the underlying cause of the cardiac arrest (e.g., blood clot, severe trauma) still needs to be addressed, and patients often face post-cardiac arrest syndrome, in which the brain, heart, and other organs have been starved of oxygen for several minutes or longer, causing inflammation and potential organ dysfunction. The 30-day survival rate therefore reflects the performance of the full emergency care pathway, including both pre-hospital care and the hospital stage.

For each outcome, we use two patient groups: (1) the conventional group covering all cardiac arrest cases, and (2) the Utstein patient group, which identifies cardiac arrests with witnessed onset, an initial shockable rhythm, and other favourable prognostic factors. The Utstein subgroup provides a more homogeneous patient population for comparison and is widely used in international cardiac arrest research. Both measures offer complementary perspectives on the relationship between emergency response performance and patient outcomes.

We focus on cardiac arrest episodes because they constitute medical emergencies in which individuals cannot predict or control the timing of the event. The random nature of cardiac arrest onset directly supports the identifying assumption that patients cannot manipulate their timing of demand around the policy cutoff.

Population mortality. We obtain weekly mortality data at local authority level from the Office for National Statistics (2023) over 2018–2023, covering all regions for which we have ambulance service performance data. Mortality data are disaggregated by location of death (care home,

home, hospice, hospital, other communal establishment, elsewhere) and by cause (all causes or COVID-19). We convert these counts into mortality rates adjusted by local population, measured as deaths per 100,000. These data allow us to present the its geographical correlation with ambulance response times.

3.2 Summary Statistics

Table 1 reports descriptive statistics for the key variables used in the analysis. Because the underlying datasets are observed at different units and frequencies, for example, local authority by week for ambulance response times and mortality, NHS Foundation Trust by day for handover, discharge, and bed capacity, and regional ambulance trust by month for cardiac arrest outcomes, therefore, the panels differ in their numbers of observations.

Panel A documents substantial variation in ambulance response times across emergency categories. Consistent with the national response targets, Category 1 (C1) calls are the most life-threatening emergencies which have the shortest mean response time of 8.54 minutes, while less urgent calls experience progressively longer delays, reaching an average of 140.32 minutes for C4 cases. The wide standard deviations, especially for lower-priority categories, indicate large temporal and regional heterogeneity in system pressure. For Incident volumes across different types, C2 calls dominate the service workload, with a mean of around 1,973 per week per local authority, and most of patients are classified as C2 and C3 patients.

Panel B summarises hospital handover delays. On average, 11.6 patients per NHS Foundation Trust per day experience handovers of 30–60 minutes, while 9.8 experience handovers exceeding one hour. The corresponding time lost averages nearly 23 hours per trust per day, with a maximum exceeding 575 hours — equivalent to over 24 ambulance-days of capacity tied up in handover at a single trust on the worst day in the sample. This dispersion illustrates the severity of bottlenecks at the hospital interface during the winter period.

Panel C reports hospital discharge performance. On a typical day, each NHS Foundation Trust has on average 179 patients who no longer meet the criteria to reside in hospital, of whom 75 are discharged. The resulting daily discharge rate averages 43%, meaning that on average more than half of medically-fit-for-discharge patients remain in hospital on any given day. This is the operational margin directly targeted by the discharge fund. Long-stay patients (length of stay ≥ 21 days) account for approximately one-third of the medically-fit-for-discharge population (mean of 56 per trust per day) but have a substantially lower discharge rate of 14%, indicating that these patients who typically require more complex post-hospital care arrangements and resource allocation face

especially binding discharge constraints.

Panel D presents cardiac arrest outcomes from the NHS England Ambulance Quality Indicators. The mean ROSC rate is 26%, rising to 48% among Utstein patients who are those with witnessed arrests, an initial shockable rhythm, and other favourable prognostic factors. Thirty-day survival averages 9% for all cardiac arrest patients and 27% for Utstein cases. Because Utstein patients are defined by more controlled and favourable clinical circumstances, both ROSC and survival rates are consistently higher than in the broader cohort, where arrests often occur in more variable and complex settings associated with greater mortality risk.

Panel E reports weekly mortality rates by place of death, expressed per 100,000 population. Hospital deaths are the largest category (mean of 8.0), followed by home (5.0) and care home deaths (4.1). Total weekly mortality averages 18.5 but varies considerably across time and regions (standard deviation 6.7). COVID-19 deaths have a lower mean (1.2) but display the greatest volatility, consistent with pandemic-related spikes.

Panel F describes hospital bed capacity. On average, NHS Foundation Trusts operated 691 adult General & Acute beds and 29 adult Critical Care beds, of which 658 and 23 respectively were occupied, implying average occupancy rates of approximately 95% for General & Acute beds and 80% for Critical Care beds. The General & Acute figure is well above safe-operation thresholds 92% and underscores the extent to which hospitals were operating at full capacity throughout the winter sample (NHS England, 2022). Influenza-occupied beds averaged 10.5 (General & Acute) and 0.8 (Critical Care) per trust per day, with large standard deviations reflecting the seasonal concentration of flu pressure.

Together, these summary statistics describe a system operating close to capacity along every observable margin. Our data shows that in England NHS system, ambulance response times elevated relative to national targets, hospital bed occupancy averaging around 95%, and daily discharge rates well below what would be required to clear the backlog of medically-fit-for-discharge patients. The empirical analysis that follows exploits the policy intervention that targeted the discharge margin and traces its propagation through this multi-stage system.

Table 1: Summary Statistics

Variable	Count	Mean	Std. Dev.	Min	Median	Max
Panel A: Ambulance Response Times (Minutes) and Incidents						

Continued on next page

Table 1 continued

Variable	Count	Mean	Std. Dev.	Min	Median	Max
C1 Response Time	75,700	8.54	2.85	0.37	7.95	140.53
C2 Response Time	76,942	32.52	21.08	0.33	27.15	439.37
C3 Response Time	76,618	109.63	78.59	5.60	88.83	1408.20
C4 Response Time	61,537	140.32	144.99	0.07	96.73	3083.02
C1 Incidents	79,411	112.11	182.70	0.00	48.00	3,775.00
C2 Incidents	79,411	1,973.10	5,416.77	0.00	279.00	89,445.00
C3 Incidents	79,411	605.80	1,726.37	0.00	108.00	38,625.00
C4 Incidents	79,411	22.63	52.04	0.00	5.00	621.00
Panel B: Handover Delays						
Handover delay 30-60 mins	20,160	11.59	11.42	0	9	86
Handover delay > 60 mins	20,160	9.78	14.50	0	2	102
Time lost (hours)	20,160	22.92	45.58	0	4.16	575.56
Panel C: Hospital Discharge						
Patients not meeting criteria to reside	32,312	179.34	105.47	0	162	1,474
Patients discharged	32,312	75.46	55.56	0	65	522
Hospital discharge rate	31,705	0.43	0.20	0	0.42	1
Patients not meeting criteria to reside (LOS≥21)	32,312	56.23	44.69	0	47	407
Patients discharged (LOS≥21)	32,312	6.12	8.48	0	5	268
Hospital discharge rate (LOS≥21)	31,563	0.14	0.16	0	0.10	1.25
Panel D: Health Outcomes of Cardiac Arrest Patients						
ROSC rate	544	0.26	0.06	0	0.27	0.6
ROSC rate (Utstein)	524	0.48	0.13	0	0.49	1
Survival rate	544	0.09	0.03	0	0.09	0.4
Survival rate (Utstein)	527	0.27	0.12	0	0.27	1
Panel E: Mortality Rates (Per 100,000 Population)						
Care Home Deaths Rate	78,877	4.11	2.73	0.00	3.66	38.40
Home Deaths Rate	78,877	4.99	2.47	0.00	4.67	22.39
Hospital Deaths Rate	78,877	8.00	3.53	0.00	7.55	48.00
Hospice Deaths Rate	78,877	0.93	0.92	0.00	0.78	7.92
COVID-19 Deaths Rate	78,877	1.20	3.00	0.00	0.00	58.67
Total Deaths Rate	78,877	18.45	6.68	1.25	17.83	87.33
Panel F: Hospital Bed Capacity						

Continued on next page

Table 1 continued

Variable	Count	Mean	Std. Dev.	Min	Median	Max
Adult General & Acute beds occupied	19,177	657.98	410.05	0	587	2,396
Adult Critical Care beds occupied	19,177	23.40	23.77	0	14	130
Adult General & Acute beds open	19,177	690.89	432.10	0	615	2,463
Adult Critical Care beds open	19,177	29.34	27.22	0	18	137
Adult General & Acute bed availability rate	18,748	0.05	0.04	0	0.04	0.18
Adult Critical Care bed availability rate	18,316	0.22	0.16	0	0.20	0.55
General & Acute flu beds	19,177	10.49	23.52	0	2	422
Critical Care flu beds	19,177	0.78	2.53	0	0	57

Notes: This table reports summary statistics for main variables used in our analysis. Response times are measured in minutes, and mortality rates are expressed per 100,000 population. The discharge rate can exceed 1 in rare cases because the numerator captures the daily flow of discharges while the denominator is measured at a daily snapshot of patients no longer meeting the criteria to reside.

4 Method

This section describes how we estimate the short-run effects of the 2022–2023 *Hospital Discharge Fund* on (i) emergency care system performance and (ii) cardiac arrest patient outcomes. Our empirical strategy is based on a Regression Discontinuity in Time (RDiT) design centred at the beginning of 2023. We follow the standard RDiT framework of Hausman and Rapson (2018) and extend the analysis to embed a seasonally adjusted difference-in-discontinuities design (Grembi *et al.*, 2016; Picchetti *et al.*, 2026). We additionally provide a detailed discussion of the identifying assumptions, diagnostic tests, and robustness checks.

4.1 RDiT Setup and Estimating Equations

We begin with a local-polynomial Regression Discontinuity in Time (RDiT) design centred at the beginning of 2023. Let t denote calendar time, normalised so that $t = 0$ corresponds to the relevant cutoff, and let Y_{it} denote an outcome for unit i at time t, where units may be ambulance trusts, local authorities, or NHS acute trusts depending on the dataset. Our baseline specification is

$$Y_{it} = \alpha_i + \tau \mathbb{1}(t \geq 0) + f_-(t)\,\mathbb{1}(t < 0) + f_+(t)\,\mathbb{1}(t \geq 0) + \varepsilon_{it}, \tag{1}$$

where α_i denotes unit fixed effects, $\mathbb{1}(t \geq 0)$ is an indicator for the post-cutoff period, and $f_-(t)$

and $f_+(t)$ are local polynomial functions of time estimated separately on each side of the cutoff. The coefficient of interest, τ, captures the discontinuous change in the outcome at the cutoff.

In practice, the estimation is implemented using local polynomial regressions with a triangular kernel. Because outcomes are observed repeatedly for the same units over time, we remove unit-specific mean differences prior to estimation, which is equivalent to including unit fixed effects in equation (1).[2]

While the baseline RDiT exploits the sharp timing of implementation, a central concern in this setting is recurring winter seasonality. To address this directly, we also estimate a seasonally adjusted difference-in-discontinuities design (DiDC) specification based on year-on-year differences (Grembi *et al.*, 2016; Picchetti *et al.*, 2026). For each unit i and calendar period t, define

$$\Delta^T Y_{it} = Y_{it} - Y_{i,t-T}, \tag{2}$$

where the outcome in period t minus the outcome in the same calendar period one year earlier. T represents the number of periods in a year at the relevant frequency, which is 52 for weekly data, 12 for monthly data, and 365 for daily data. We then estimate the same local RDiT specification on the differenced series:

$$\Delta^T Y_{it} = \alpha_i + \tau_\Delta \mathbb{1}(t \geq 0) + f_-(t)\mathbb{1}(t < 0) + f_+(t)\mathbb{1}(t \geq 0) + \varepsilon_{it}. \tag{3}$$

This specification removes any seasonal component that is stable across years for a given unit, and therefore provides a more demanding comparison around the January 2023 cutoff. The coefficient τ_Δ captures whether the year-on-year change in the outcome shifts discontinuously at the start of 2023 compared with any generalized seasonal shifts observed. A statistically significant estimate implies that outcomes improved or deteriorated more sharply at that cutoff than would be predicted by the pre-cutoff path of year-on-year differences. We treat this seasonally adjusted specification as our main empirical design because it offers a conservative way to net out any recurring winter patterns that are common across years.

Equations (1) and (3) are applied consistently across datasets of different temporal frequencies: weekly data for ambulance response times, monthly data for ROSC and 30-day survival rates, and daily data for hospital discharge, handover delays, adult hospital bed availability, and in-

[2]Because outcomes are observed repeatedly for each ambulance trust or local area, we remove unit-specific mean differences to isolate within-unit variation over time. In practice, this is equivalent to including unit fixed effects in equation (1).

fluenza bed occupancy. In each case, the running variable is normalised relative to the corresponding cutoff.

Our main estimates use mean squared error (MSE)-optimal bandwidth selection and robust bias-corrected confidence intervals (Calonico *et al.*, 2014, 2019). Following standard practice in the RD literature, we use a local linear specification ($p = 1$) in the main analysis.[3] Standard errors are clustered at the local-unit level to account for serial correlation and unobserved shocks within ambulance trusts, local authorities, or acute trusts over time.

4.2 Identifying Assumptions and Diagnostic Checks

The RDiT design identifies a causal effect under the assumption that, absent the policy intervention, the outcome would have evolved smoothly through the cutoff. In our setting, this requires that no other factor affecting emergency care performance or patient outcomes changed discontinuously at the beginning of 2023 in a way that coincides with the implementation of the discharge fund.

More specifically, the identification strategy relies on three conditions.

1. **Continuity of counterfactual outcome paths.** In the absence of the discharge funding intervention, the conditional expectation of each outcome would evolve smoothly through the cutoff. Any discontinuity at $t = 0$ is therefore attributed to the intervention rather than to an unrelated structural break.

2. **No simultaneous discontinuity in major confounders.** Other determinants of emergency care performance and cardiac arrest outcomes should not change discretely at the same threshold. This is particularly important in a time-based RD design, where seasonal pressures and contemporaneous institutional shocks are natural concerns.

3. **No precise manipulation of the running variable.** Agents cannot sort around the cutoff in calendar time. In our context, patients cannot time life-threatening emergencies; providers did not pre-announce the exact timing of implementation in a way that allows precise manipulation at the time boundary.

We conduct several diagnostic checks to assess these assumptions. First, we estimate placebo RDiT specifications centred at January 2022 and January 2024. If the discontinuity observed in

[3] For ambulance response times, we report specifications using a quadratic polynomial, as the pre- and post-cutoff trends display some curvature in the visual evidence.

2023 merely reflects recurring winter seasonality, similar breaks should appear at those placebo thresholds. The absence of comparable placebo effects would therefore strengthen the interpretation that the 2023 discontinuity is linked to the discharge funding intervention.

Second, we examine influenza-related hospital pressure around the cutoff using data on general-and-acute and critical-care beds occupied by patients with laboratory-confirmed influenza. Influenza is an important potential confounder because it may independently raise emergency demand and strain hospital capacity. If influenza pressure evolves smoothly through the cutoff, this reduces the concern that the estimated treatment effects are driven by a concurrent change in underlying disease burden.

Third, we re-estimate the main specifications for the East of England Ambulance Service, the only ambulance region not directly affected by industrial strike action during the relevant period. This subsample analysis provides a useful diagnostic for whether contemporaneous labour disruptions could be driving the estimated discontinuities.

4.3 Discussion of the Policy Cutoff

In a Regression Discontinuity in Time (RDiT) design, the running variable is calendar time rather than a cross-sectional score (Hausman and Rapson, 2018). Identification therefore depends critically on locating the time threshold at which treatment exposure changes sharply. In time-based settings, this choice is especially important because institutional implementation may lag behind formal policy announcements, and multiple seasonal or operational events may occur within a short window. We therefore begin by discussing the appropriate cutoff for the 2022–2023 discharge funding intervention.

In our setting, the NHS *Hospital Discharge Fund* was allocated to local systems in early December 2022. At first glance, December 2022 might appear to be the natural treatment onset. However, both institutional considerations and the observed timing of system responses suggest that December is not the relevant implementation threshold. Instead, January 2023 provides the cleaner and more empirically appropriate cutoff. We explain this choice in detail below, and we also discuss how we align the cutoff across datasets observed at different temporal frequencies.

Why December 2022 is not an effective cutoff. Although funding allocations were confirmed in early December, converting those allocations into effective discharge capacity required time. The fund was primarily intended to support the procurement of step-down care beds, the arrangement of home-care packages, and the expansion of discharge coordination teams. In practice, these ac-

tivities are typically implemented through weekly or monthly contracting, staffing, and rostering cycles. As a result, December 2022 should be viewed as a transitional period in which funding had been announced but had not yet been fully translated into operational discharge capacity across local systems. Treating December as the cutoff would therefore risk mixing partially treated and untreated observations, which would attenuate estimated effects and weaken the interpretation of the discontinuity. For checking the robustness of policy timing choices, we also report estimates using Donut RDiT specifications which we discussed in Section 4.4.4.

Why January 2023 is the relevant implementation cutoff. By the beginning of January 2023, local systems had begun deploying the newly funded discharge capacity at scale. As shown in Figure B.3, administrative data indicate that by the end of the first reporting period on 4 January 2023, approximately £90 million—around 18% of the total fund—had already been disbursed. This pattern suggests that implementation accelerated rapidly during the first full week of January. Consistent with this interpretation, the first-stage results in later Section **??** show a clear and sustained improvement in ambulance response times beginning in the first week of 2023, with no comparable discontinuity in December 2022. Taken together, the institutional timing and the observed first-stage response imply a short implementation lag between fund allocation and operational impact. Since RDiT identifies a local treatment effect at the point where treatment intensity changes discontinuously, the appropriate cutoff is the point at which the intervention becomes operational at scale and performance indicators begin to respond sharply in the start of 2023.

Temporal alignment across datasets. To maintain consistency across datasets observed at different frequencies, we define frequency-specific cutoffs that correspond to the same underlying implementation threshold. For monthly outcomes, such as ROSC and 30-day survival rates, the cutoff is January 2023. For weekly outcomes, such as ambulance response times and local mortality, the cutoff is the first week of 2023. For daily measures, including hospital discharge, handover delays, adult hospital bed availability, and influenza bed occupancy, we use 5 January 2023, the midweek date of the first week of 2023. These cutoffs represent the same effective implementation threshold expressed at different temporal resolutions.

4.4 Robustness and Identification Checks

4.4.1 Bandwidth and polynomial order

Local RD estimators involve a standard bias–variance trade-off through the choice of bandwidth h and polynomial order p. A narrower bandwidth places greater weight on observations closest to the cutoff and therefore reduces specification bias, but at the cost of lower precision. A wider bandwidth increases precision, but may introduce bias if the underlying time path is more curved farther from the threshold. Likewise, higher-order polynomials can fit the data more flexibly, but may also overfit local noise.

To assess the sensitivity of our results, we conduct two sets of robustness checks. First, we vary the estimation bandwidth over a symmetric grid around the MSE-optimal choice and examine the stability of the estimated discontinuity. Second, we re-estimate the models using polynomial orders $p \in \{1,2,3\}$. The local linear specification remains our preferred benchmark because it is standard in the RD literature and avoids unnecessary overfitting, but the broader set of estimates provides a transparent assessment of robustness.

4.4.2 Balanced data and weighted estimation

The cardiac arrest dataset is structured at the ambulance trust-by-month level. A practical issue is that the panel is not fully balanced because observations for the Isle of Wight Ambulance Trust are occasionally missing. This occurs because the number of cardiac arrest cases in that trust is very small, so rates such as ROSC or 30-day survival are undefined in months with no eligible patients.

In the main specification, we therefore retain the unbalanced panel and weight each observation by the number of cardiac arrest patients in the corresponding trust-month cell. This weighting scheme reduces the influence of cells based on very small numbers of patients and mitigates the extent to which missing outcomes from a very small trust affect the aggregate estimates. As a robustness check, we also report results from (i) a balanced panel that excludes the Isle of Wight Ambulance Trust and (ii) the unbalanced panel estimated without patient-count weights.

4.4.3 Clustering of standard errors

In the main specification, standard errors are clustered at the local-unit level—that is, at the ambulance trust, local authority, or NHS acute trust level depending on the dataset. This clustering choice allows for arbitrary serial correlation and heteroskedasticity within local units over time,

which is important because treatment exposure, operational conditions, and measurement errors are likely to be correlated within the same unit across adjacent periods. As an additional robustness check, we also report alternative inference based on clustering by time and two-way clustering by unit and time where feasible.

4.4.4 Donut RDiT

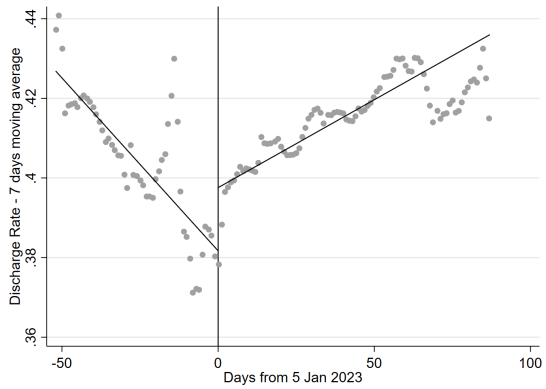
Observations immediately around the cutoff may be affected by short-lived implementation frictions or other transitory disturbances. If so, the estimated discontinuity could be driven disproportionately by a small number of near-threshold observations rather than by a broader shift in the local outcome path. To address this, we implement a "donut" RDiT that drops observations in a symmetric window around the cutoff and re-estimates the model on the remaining data (Barreca *et al.*, 2011; Noack and Rothe, 2023). Given our relatively narrow optimal windows, we remove ± 1 month around $t = 0$ for monthly outcomes (and the analogous $\pm$ window for weekly performance when reported). If the estimated discontinuity reflects a genuine policy effect rather than a handful of atypical observations at the threshold, the sign and broad magnitude of the estimates should remain similar, albeit with wider confidence intervals because of the reduced sample.

5 Results

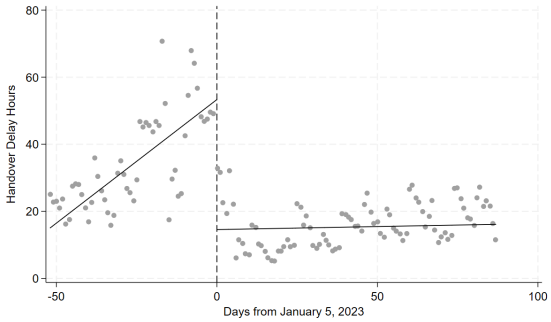
5.1 Policy Impact on System Performance: Discharge, Handovers, and Response Times

In this section, we first examine whether the 2022–2023 *Hospital Discharge Fund* successfully improved hospital discharge capacity, as intended. The main purpose of the Discharge Fund is to improve hospital discharge capacity, releasing more patients who no longer meet the criteria to reside in hospital to step-down care or home care, in order to reduce congestion for waiting patients. If discharge improves, resources including beds and staff are freed up, ambulances can hand patients over more quickly, and vehicles and paramedics are released to respond to the next call. We therefore examine the three stages of this chain in turn: hospital discharge, ambulance handover delays, and ambulance response times. Figure 1 presents the discontinuity at the policy cutoff for all three outcomes; the corresponding estimates, together with placebo cutoffs at January 2022 and January 2024 and a difference-in-discontinuities (DiDC) specification, are reported in Appendix Tables B.1–B.3.

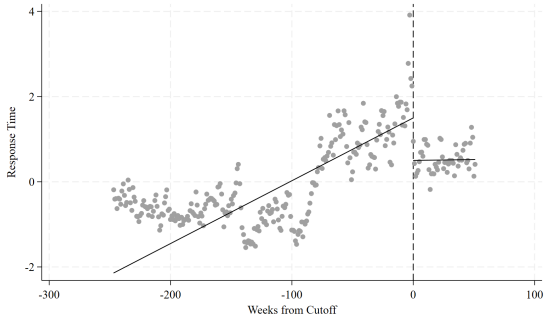
Figure 1: Discontinuities in System Performance at the Policy Cutoff



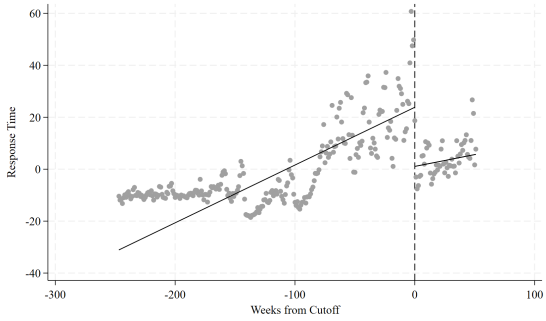
(a) Hospital discharge rate (7-day MA)



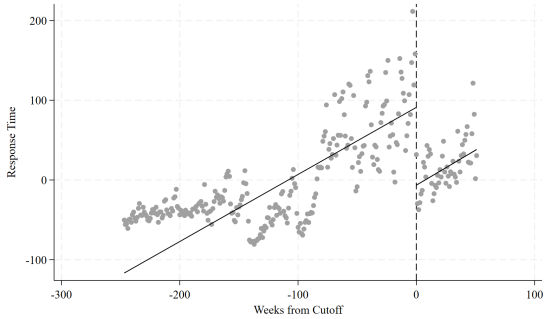
(b) Daily handover delay hours



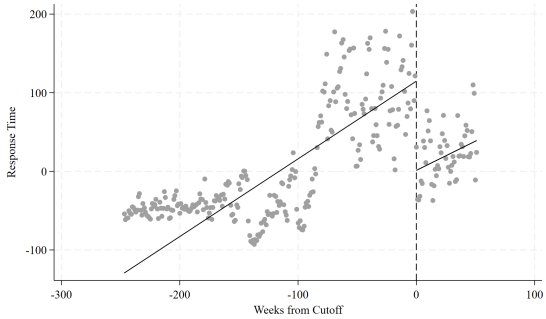
(c) Response times, C1



(d) Response times, C2



(e) Response times, C3



(f) Response times, C4

Notes: Panels (a) and (b) use daily trust-level data around the 5 January 2023 cutoff; panels (c)–(f) use weekly local-authority data with the cutoff at the first week of 2023. Fitted lines are local linear fits on each side of the cutoff, consistent with the estimation in Appendix Tables B.1–B.3. Local quadratic versions of panels (c)–(f) are shown in Appendix Figure B.10.

Hospital discharge

Panel (a) of Figure 1 presents the discontinuity in the hospital discharge rate around the 5 January 2023 cutoff. After the cutoff, the discharge rate improves steadily and recovers toward its pre-winter level, rather than exhibiting the brief surge-and-collapse pattern seen around Christmas.[4] This persistent post-January recovery is more consistent with a structural improvement in discharge capacity, whereby the funding enabled trusts to expand discharge pathways, arrange additional step-down care, and strengthen discharge coordination, and it supports our choice of 5 January 2023 as the relevant cutoff for the daily analysis.

Appendix Figure B.8 assesses whether this pattern could reflect recurring winter seasonality by comparing winters 2021–22 and 2022–23. Discharge rates in 2022–23 are generally lower, indicating substantially greater pressure, yet after early January the 2022–23 series recovers toward its pre-winter level while the 2021–22 series continues to weaken. The post-January rebound in 2022–23 therefore cannot be explained solely by seasonality. Consistent with this, Appendix Table B.1 shows a discontinuity of 2.8 percentage points in the overall discharge rate at the 5 January 2023 cutoff ($p < 0.01$), with no comparable discontinuity at the placebo cutoffs in January 2022 or January 2024. For patients with a length of stay of at least 21 days, the point estimate is positive but imprecise, suggesting that the marginal discharges enabled by the fund were concentrated among patients whose onward care could be arranged quickly, rather than long-stay patients facing binding social-care capacity constraints.

Ambulance handover delays

When emergency departments operate near capacity, ambulances have to wait at hospital and remain clinically responsible for their patients; long handovers constrain the supply of ambulances in service and therefore lengthen response times.[5]

[4] There is a sharp but short-lived spike around 22 December 2022, likely reflecting a tactical pre-Christmas discharge effort, with trusts temporarily accelerating discharges before the holiday period. This increase quickly reverses during the Christmas–New Year period, when discharge rates fall to their lowest level in the sample, indicating that the late-December spike does not represent a sustained improvement in discharge capacity. A comparison with winter 2021–22 (Appendix Figure B.8) shows a similar temporary Christmas spike in both winters, confirming that this is a recurring holiday effect rather than a consequence of the 2022–23 funding.

[5] The handover time is the period between an ambulance's arrival at a hospital and the formal transfer of the patient's care to hospital staff; a handover delay occurs when this process exceeds 15 minutes. Descriptively, the share of arrivals with handovers exceeding 30 minutes peaked at nearly 45% in late December 2022 and fell sharply thereafter, stabilising at roughly 20%–30% through the first quarter of 2023 (Appendix Figure B.22).

Panel (b) of Figure 1 provides the regression discontinuity plot of handover delay hours at the daily trust level. The fitted lines show a sharp structural break at the 5 January 2023 cutoff: total delay hours drop abruptly, consistent with a substantial release of ambulance capacity. Appendix Table B.2 quantifies this break: hours lost to handover delays fall by 0.28 hours (about 17 minutes) per arrival, and the share of arrivals delayed more than 60 minutes falls by 5.4 percentage points ($p < 0.01$), while the share delayed more than 30 minutes falls by a smaller and marginally insignificant 3.7 percentage points. The improvement is thus concentrated in the right tail of extreme delays—precisely the delays most costly for ambulance availability.[6]

Ambulance response times

Given the improvements in hospital discharge and handover delays, we would expect a corresponding improvement in ambulance response times, as ambulances could hand over patients more promptly and respond to waiting patients more efficiently. Panels (c)–(f) of Figure 1 display regression discontinuity plots for each emergency category (C1–C4) around the first week of 2023. A clear discontinuous drop is visible across all categories immediately after the threshold: prior to the cutoff, response times exhibited gradually increasing trends, particularly during the COVID-19 period, while the post-cutoff period shows sustained lower response times across all categories.[7]

Appendix Table B.3 reports the estimated discontinuities. For life-threatening emergencies (Category 1), the weekly average response time falls by approximately 2.5 minutes, from around 11 to around 9 minutes. For emergency but non-life-threatening incidents (Category 2), response times decline by roughly 29 minutes, from about 70 to about 40 minutes. For lower-priority cases (Categories 3 and 4), which face the longest delays, average response times drop by roughly 145–160 minutes from pre-cutoff levels of 250–300 minutes. All estimates are significant at the 1% level. The magnitude and uniformity of these improvements across categories suggest a system-wide operational shift rather than category-specific interventions, consistent with the structural relief provided by increased discharge capacity.

Because January discontinuities in response times could partly reflect mean reversion from the December winter peak, Appendix Table B.3 also reports a placebo cutoff at January 2022 and the DiDC specification. A smaller January drop is present in the placebo year, confirming a sea-

[6] A daily event-study of handover delay hours relative to 1 January 2023 (Appendix Figure B.23) shows that the reduction is sharp, statistically significant within the first full week of 2023 (2–8 January), and persistently negative thereafter—coinciding precisely with the week in which the improvement in ambulance response times begins in our weekly data.

[7] The monthly time series over the full study period is shown in Appendix Figure B.4.

sonal component; however, the 2023 discontinuities are four to five times larger, and remain large and significant in the DiDC specification, which nets out the seasonal January pattern by differencing each local authority against the same week of the previous year. The discontinuity at the start of 2023 therefore cannot be attributed to routine winter seasonality. Collectively, these results indicate that the discharge funding was followed by a sharp and sustained reduction in handover delays, which released ambulance capacity and shortened response times to new calls. This mechanism provides a direct operational explanation for the improvements in patient outcomes we examine next.

5.2 Main Effects: Patient Outcomes

This section examines whether the sharp improvement in ambulance response times identified in Section 5.1 translated into measurable changes in patient health outcomes. We first analyse a direct pre-hospital outcome, *Return of Spontaneous Circulation* (ROSC) *rate* on Arrival at Hospital for cardiac arrest patients, which measures the system-wide returns to the shorter ambulance response times. The ROSC rate reflects the effectiveness of pre-hospital intervention, particularly the timeliness of cardiopulmonary resuscitation (CPR) and defibrillation, and therefore serves as a physiological indicator of whether faster response directly improved the chances of initial survival (Neumar *et al.*, 2015; Nolan *et al.*, 2010).

We then turn to *30-day survival rate*, a broader measure that captures both pre-hospital and in-hospital components of emergency care. Faster ambulance response allows patients to receive life-saving treatment earlier at the scene, while reduced handover delays enable quicker transfer to hospital care. Together, these mechanisms reflect how the overall performance of the emergency care system translates into sustained survival improvements beyond the initial incident.

For each outcome, we present visual evidence of the discontinuity around the January 2023 cutoff, followed by the estimated treatment effects from our RDiT models. We also report a series of robustness checks, including placebo cutoffs, donut specifications, bandwidth choices, and polynomial order sensitivity analyses, to ensure that our findings are not driven by specification choices or short-term confounders in the next section.

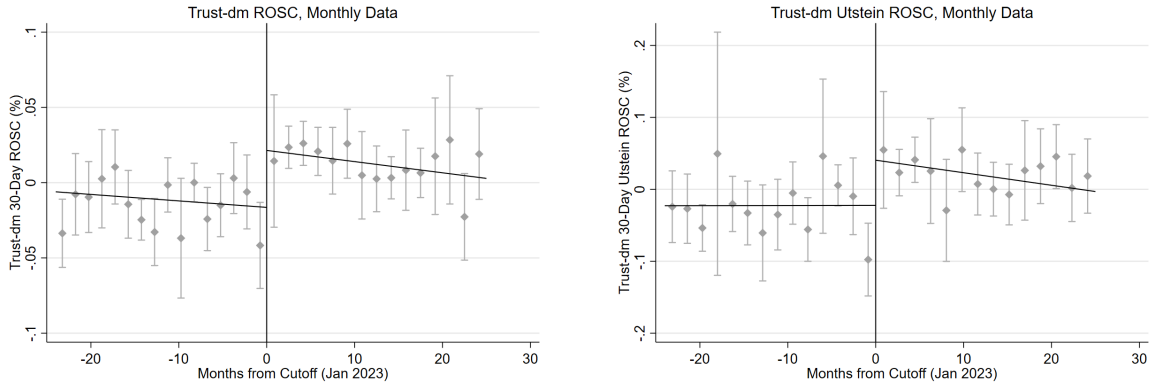
5.2.1 Immediate pre-hospital outcome: return of spontaneous circulation rate on arrival at hospital

We begin our analysis of health outcomes with the Return of Spontaneous Circulation (ROSC) rate, a pre-hospital indicator of successful resuscitation following cardiac arrest. It stands for the mo-

ment the ambulance crew's resuscitation treatment (e.g., CPR or defibrillation) is successful, and the patient's heart, which had stopped, starts beating on its own again. This ROSC rate measures the proportion of patients who regained a palpable pulse upon arrival at hospital among those for whom resuscitation was attempted. This outcome directly reflects the timeliness and effectiveness of pre-hospital emergency response, especially for cardiac arrests, where the probability of survival declines sharply with each minute of delay in initiating CPR or defibrillation.

Figure 2a and Figure 2b present regression discontinuity plots of the ROSC rate and the Utstein ROSC rate.[8] In both panels, there is a clear upward shift at the cutoff corresponding to the beginning of 2023. This discontinuity coincides precisely with the improvement in ambulance response times documented in Section 5.1, this aligns with that in theory, faster ambulance response could translate into higher rates of successful resuscitation at the scene.

Figure 2: Regression Discontinuity in Time: ROSC Rate



(a) ROSC Rate

(b) ROSC Rate for Utstein Patients

The RDiT estimates reported in Table 2 confirm the visual evidence. Across trusts, the ROSC rate increased by approximately 4.5 percentage points ($p<0.01$) following the January 2023 cutoff, while the Utstein ROSC rate experienced a much larger and statistically significant improvement of 15.2 percentage points. Placebo tests using the same specification at both a January 2022 and a January 2024 cutoff yield statistically insignificant estimates in both panels, and the DiDC estimates in specification (4) further validate that the observed discontinuity is unique to 2023.[9] The DiDC results provide a seasonally-adjusted estimate for the treatment effect after netting out seasonal patterns using the year-on-year differenced outcome: the point estimate for overall ROSC is 3.8 percentage points ($p<0.10$), while the Utstein ROSC effect strengthens to 19.3 percentage

[8] The Utstein comparator group includes bystander-witnessed, non-traumatic cardiac arrests with a shockable rhythm, providing a clinically comparable sample across time (Shin *et al.*, 2021).

[9] See Figure B.11 and Figure B.13 for a visual plot of the RDD around the placebo cutoff.

points (p<0.01), suggesting that even though the seasonal component attenuates the main estimate, the effect is still significant and positive. Figure B.15 presents the corresponding RD plot, where a clear discontinuity is visible at the January cutoff. The effect remains positive but not significant when we restrict the sample to the East of England Ambulance Service, which was the region not affected by the industrial strikes over this period. This is reasonable as the sample size is decreased a lot by restricting the sample into a specific trust.

Table 2: Main RDD Results: ROSC

	(1) Main RDD	(2) Placebo 2022	(3) Placebo 2024	(4) DiDC	(5) East England
Panel A: ROSC rate					
Treatment effect	0.045***	0.019	0.008	0.038*	0.043
Standard error	(0.016)	(0.019)	(0.024)	(0.031)	(0.048)
P-value	[0.0004]	[0.2575]	[0.5892]	[0.0601]	[0.3410]
Observations	97	187	98	76	7
Mean outcome	0.005	-0.010	0.010	0.021	0.052
Bandwidth (months)	4.48	8.68	4.78	3.45	3.27
Panel B: ROSC rate (Utstein)					
Treatment effect	0.152***	0.005	-0.015	0.193***	0.350
Standard error	(0.033)	(0.046)	(0.039)	(0.070)	(0.358)
P-value	[0.0000]	[0.8716]	[0.7813]	[0.0007]	[0.1691]
Observations	94	140	154	73	5
Mean outcome	0.009	-0.020	0.018	0.028	0.096
Bandwidth (months)	4.43	6.61	7.71	3.13	2.35
Cutoff Year	2023	2022	2024	2023	2023
East of England Only	No	No	No	No	Yes

Notes: Conventional point estimate; (robust bias-corrected SE); [robust bias-corrected p-value]. Col. (1): RDiT specification at Jan 2023 cutoff; cols. (2)–(3): placebo cutoffs at Jan 2022 and Jan 2024; col. (4): main specification with year-on-year differenced outcome at Jan 2023 cutoff; col. (5): East of England only for specification (4). Unbalanced panel (11 trusts) with `number of cardiac arrest patients` weights except col. (5). Running variable: months from January 2023. Local linear estimator, triangular kernel, MSE-optimal bandwidth. Standard errors and p-values from Calonico *et al.* (2014) robust bias-corrected inference; clustered by ambulance trust (col. 5: heteroskedasticity-robust). *** p<0.01, ** p<0.05, * p<0.10.

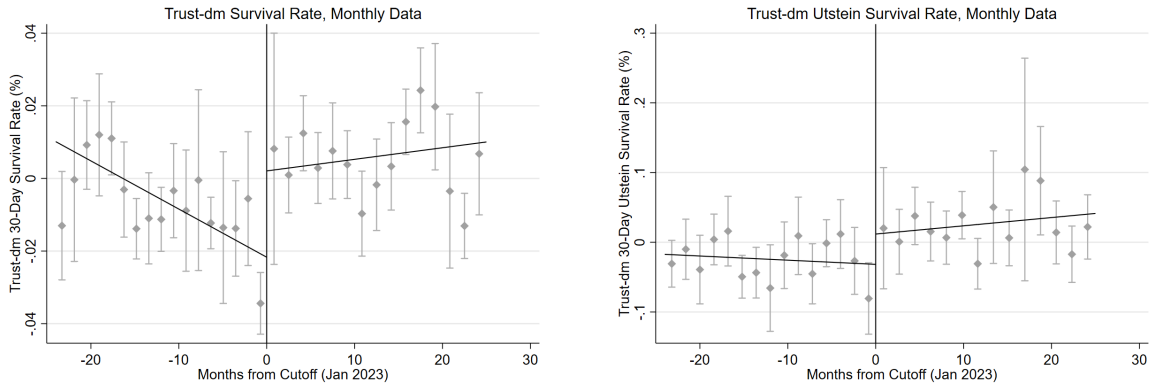
Taken together, the ROSC results provide strong evidence of the causal mechanism linking system performance and survival. Faster ambulance response times increased the likelihood that patients received timely CPR or defibrillation, allowing spontaneous circulation to be restored before hospital arrival. This immediate gain forms the first link in the broader chain through which improved emergency care system performance translated into higher survival for cardiac arrest patients, which we examine next.

5.2.2 Clinical outcome: survival rate at 30 days

We proceed by examining the 30-day survival rate following cardiac arrest as the primary outcome of interest. This measure provides a direct and time-sensitive indicator of full emergency system performance, as patient survival after out-of-hospital cardiac arrest depends critically on both early resuscitation interventions and timely hospital care.

Figure 3a and Figure 3b present the regression discontinuity plots for the overall 30-day survival rate and the Utstein comparator group, respectively. In both cases, we observe a clear and discrete upward shift in survival immediately after the January 2023 cutoff, consistent with the timing of the sharp reduction in ambulance response times documented earlier (see Section 5.1). The increase is more visible in the general survival rate than in the Utstein comparator group. This pattern aligns with the clinical nature of each outcome: Utstein patients whose cardiac arrests are witnessed and typically receive early bystander intervention face lower death risks and are therefore less sensitive to additional minutes of ambulance delay. In contrast, the broader patient population includes many unwitnessed arrests in which survival depends critically on the speed of professional response. The observed discontinuity thus provides evidence that the improvement in full emergency care system performance at the start of 2023 translated into better survival outcomes, particularly among those most reliant on timely emergency care.

Figure 3: Regression Discontinuity in Time: Survival Rate at 30 Days



(a) Survival Rate at 30 Days

(b) Survival Rate at 30 Days for Utstein Patients

Table 3 summarizes the formal RDiT estimates. For the overall 30-day survival rate, the main RDD coefficient indicates an increase of 5.2 percentage points ($p<0.01$) following the 2023 cutoff, implying a statistically significant improvement in patient outcomes. This magnitude, even though modest, is still meaningful given the low baseline survival rate in out-of-hospital cardiac

arrest cases. For the Utstein comparator group, the estimated effect is 8.0 percentage points and statistically significant at the 5% level ($p = 0.012$), reinforcing the finding that the policy discontinuity translated into tangible gains in patient survival.

The placebo tests help adjudicate whether the 2023 jump reflects a genuine policy effect or a recurring seasonal pattern. The January 2022 placebo yields a significant coefficient of 2.1 percentage points for overall survival, and the January 2024 placebo similarly shows a small but significant estimate of 0.8 percentage points. However, both are considerably smaller than the main 2023 estimate, and crucially, neither Utstein placebo is statistically significant, suggesting that the large Utstein survival gain at the 2023 cutoff is not driven by seasonal variation, reporting artefacts, or unrelated winter shocks.[10]

The DiDC estimate compares the discontinuities across the two years to net out any common seasonal patterns. For the overall 30-day survival rate, the year-on-year differenced estimate remains significantly positive at 2.4 percentage points (p<0.01), confirming that the 2023 improvement exceeds what can be attributed to regular seasonal fluctuations. The Utstein DiDC estimate is 5.4 percentage points but not statistically significant ($p = 0.256$), which likely reflects the added noise from differencing rather than an absence of the underlying effect, given the strong main result in Panel B. In Figure B.16, we still can observe an evident jump after the cutoff, even though it is less apparent for the Utstein group. We also present the estimates using only East of England data, where no strike action occurred during December 2022 or January 2023. The estimates are not statistically significant, 4.3 percentage points for overall survival and -6.2 percentage points for Utstein.

5.3 Robustness

In this section, we present a series of robustness checks to validate the main DiDC estimates for the ROSC and survival outcomes. We first test for various bandwidth and polynomial order specifications to ensure that our results are not driven by particular choices in the RDD estimation. We then additionally present he donut RDD visual plots to further rule out the possibility that the observed discontinuity is driven by short-term confounders around the cutoff. Moreover, we also present the DiDC estimates with different restricted samples such as the balanced panel which excludes the trusts with missing data in some months, and the sample with full data for the whole period, to check whether the results are robust to the sample selection. We also provide the DiDC estimates with different clustering levels at the time level and the trust × time level. Finally, we

[10] See Figure B.12 and Figure B.14 for a visual plot of the RDD around the placebo cutoff.

Table 3: Main RDD Results: 30-Day Survival

	(1) Main 2023	(2) Placebo 2022	(3) Placebo 2024	(4) DiDC	(5) East England
Panel A: 30-day survival					
Treatment effect	0.052***	0.021***	0.008**	0.024***	0.043
	(0.008)	(0.009)	(0.006)	(0.012)	(0.078)
	[0.0000]	[0.0088]	[0.0361]	[0.0052]	[0.3218]
Observations	76	143	182	97	5
Mean outcome	-0.006	-0.005	0.009	0.002	0.031
Bandwidth (months)	3.26	6.48	8.29	4.16	2.29
Panel B: 30-day survival (Utstein)					
Treatment effect	0.080**	0.027	-0.016	0.054	-0.062
	(0.040)	(0.035)	(0.030)	(0.066)	(0.229)
	[0.0116]	[0.3611]	[0.7161]	[0.2555]	[0.7592]
Observations	135	118	137	113	7
Mean outcome	0.000	-0.032	0.030	0.021	0.095
Bandwidth (months)	6.59	5.86	6.53	5.73	3.49
Cutoff Year	2023	2022	2024	2023	2023
East of England Only	No	No	No	No	Yes

Notes: Conventional point estimate; (robust bias-corrected SE); [robust bias-corrected p-value]. See Table 2 for column definitions. Outcome is 30-day survival rate. Utstein comparator: cardiac arrests with shockable initial rhythm. *** p<0.01, ** p<0.05, * p<0.10.

test whether the discontinuity among all outcomes is driven by the same timing of the variations of the winter influenza by checking the RDD plots of the number of beds occupied by influenza patients, which is a key seasonal factor that may affect the demand of emergency care and the baseline health outcomes of patients.

5.3.1 Bandwidth and polynomial order selection

The above RDD treatment effects are the differences of fitted values at the cutoff time estimated by pre-cutoff and post-cutoff data respectively, while they may be sensitive to the choice of bandwidth and polynomial order of the fitting curve. We test this sensitivity and present it in the Table B.4, which includes all estimates of DiDC with different settings. Panel A varies the local bandwidth around the optimal value while holding the polynomial order fixed at one. The estimated effects remain consistently positive and statistically significant across a wide range of bandwidths. For the ROSC rate, coefficients decline gradually in magnitude from roughly 0.066 to 0.023 as the bandwidth widens, but the estimates remain highly significant throughout. For the Utstein sample, the point estimates follow a similar pattern, which is larger at smaller bandwidths. This behaviour is typical of RDD settings, as narrower bandwidths capture the sharp local change near the cutoff, while wider ranges incorporate more noise from surrounding periods. Panel B assesses sensitivity to polynomial specification using the optimal bandwidth for each outcome, where we change the polynomial order from 1 to 3. The estimated effect remains stable and significant across all polynomial orders, with larger magnitudes at higher-order fits. For the sake of avoiding the potential overfitting, we finally choose the linear function to show the main DiDC estimates reported in Section 5.2, which gives a more conservative estimate.

Sensitivity analyses in Table B.5 broadly support the baseline estimates. In Panel A, the full-sample estimates are positive and statistically significant at all bandwidths below 70% and from 100% upward, with coefficients declining smoothly from 0.028 at 50% to 0.015 at 150% as the local window widens and the treatment contrast dilutes. The 80–90% bandwidth range yields insignificant estimates despite positive point estimates of similar magnitude; we attribute this to the discrete structure of the monthly running variable, whereby small bandwidth reductions correspond to the removal of whole monthly observations rather than marginal data points, producing step-changes in effective sample composition rather than smooth local variation. The Utstein subsample yields uniformly positive point estimates across all bandwidths (ranging from 0.042 to 0.063) but does not achieve statistical significance at conventional levels. Panel B examines polynomial order sensitivity at the MSE-optimal bandwidth for each order. The $p = 1$ and $p = 2$ specifications yield consistent, statistically significant estimates for the full sample (0.024 and 0.034 respectively),

providing confidence in the baseline local linear result. Higher-order specifications, however, are uninformative in this setting: with an MSE-optimal bandwidth of approximately four months, the local estimation window contains too few monthly observations to support the identification of cubic variations. This pattern is consistent with the caution against high-order polynomial RD specifications in short windows raised by Gelman and Imbens (2019), and reinforces the appropriateness of the local linear baseline. For the Utstein subsample, estimates remain positive across all polynomial orders but are imprecisely estimated throughout, again reflecting the limited power available within this subgroup. Overall, the sensitivity analyses suggest that the main DiDC estimates for ROSC rate and 30-day survival are not supposed to be driven by specific bandwidth or polynomial choices, and the treatment effects are generally robust to a wide range of bandwidth and polynomial choices, with the local linear specification providing a conservative and reliable estimate of the treatment effect.

5.3.2 Donut RDD

To eliminate the concern that our RDD estimates are just driven by other short term unobservable variations, we perform a Donut RDD analysis. The idea of Donut RDD is removing the data points around the cutoff (in the "donut"), then we use the remaining data to estimate the effect. Figure B.17 and Figure B.18 present the corresponding Donut RDD analyses, in which observations within ± 1 month of the cutoff are excluded. The donut plots reveal that the positive shift in both groups persists even after removing the months closest to the cutoff, confirming our discontinuity of ROSC rates and the estimated impacts are robust.

5.3.3 Balanced data

In this section, we test the robustness of our main DiDC estimates to the use of a balanced panel and the exclusion of Isle of Wight Ambulance Trust, which has a small number of observations and contains missing values for some outcomes in several months. The main specification includes all available data from the 11 ambulance trusts, with `cardiac arrest number` weights to account for the unbalanced nature of the panel. To check whether our results are sensitive to this choice, we re-estimate the DiDC effects using (a) a balanced panel that excludes Isle of Wight and includes only trusts with complete data across all months, without applying weights; and (b) the full unbalanced panel but without applying any weights.

Table B.6 presents the results. Panel A shows that when we restrict to a balanced panel excluding Isle of Wight, the estimated treatment effects remain positive and statistically significant for

both ROSC rates and 30-day survival rates, which are similar to the estimates in the main specification. This implies that the weight adjustment in the main specification minimize the potential bias from the unbalanced panel, and we do not have to drop the information from Isle of Wight to get consistent estimates. Panel B includes all available data without weighting, which yields similar patterns but with larger standard errors and less precise estimates, again consistent with the loss of balance and increased noise because of including Isle of Wight data. Overall, these robustness checks suggest that our main findings are not driven by specific sample restrictions or weighting choices, and that the observed improvements in ROSC rates and survival are robust to alternative specifications.

5.3.4 Clustering of standard errors

Another concern is that the standard errors in our main specification may be selected by specific forms of clustering in the data. In our main analysis, we cluster standard errors at the ambulance trust level, which accounts for within-trust correlation over time. However, given the monthly structure of our running variable and the potential for common shocks affecting all trusts in a given month (e.g., weather conditions, policy changes), it is also important to consider clustering at the calendar month level. Table B.7 presents the results of DiDC estimates with standard errors clustered by calendar month and Month×Trust, respectively. When clustering by calendar month, the estimated treatment effects remain positive and statistically significant for both ROSC rates and 30-day survival rates, with similar magnitudes to the main specification. This suggests that our findings are not driven by common shocks affecting all trusts in specific months. When clustering by Month×Trust, which is a more conservative approach that accounts for both within-trust and within-month correlation, the estimated effects are still positive but become less precise. Overall, these results indicate that while our main findings are robust to different clustering choices, the precision of the estimates can be sensitive to the level of clustering, which is an important consideration for interpreting the statistical significance of our results.

5.3.5 Identification Check: Winter Influenza Pressure

A key identifying assumption in our design is that the discontinuity observed in cardiac arrest patient outcomes at the start of 2023 was not driven by concurrent changes in other health system pressures that could independently affect patient outcomes. One major confounding concern is the intensity of the winter influenza season. Influenza outbreaks typically peak in December and January and are associated with substantial seasonal fluctuations in hospital demand and mortal-

ity. If influenza prevalence dropped sharply at the same time as the improvement in ambulance response times, our estimated effects on patient outcomes could in part reflect a shift in background health risks rather than the causal impact of emergency service performance.

To examine this, we use daily data from NHS England on hospital bed occupancy by patients with laboratory-confirmed influenza. We distinguish between General and Acute (G&A) beds, which capture the broader inpatient caseload, and Critical Care (CC) beds, which represent the most severe flu cases requiring intensive treatment. Figures B.20a and B.20b present regression discontinuity plots of these two measures around the 1st January 2023 cutoff.

Both figures show a clear seasonal decline in influenza bed occupancy after the late-December peak, consistent with the typical winter pattern. However, this decline occurs smoothly over time, without any significantly visible break or discontinuity at the cutoff. In both G&A and critical care settings, the number of flu-related admissions decreases gradually, indicating that the underlying influenza pressure on the healthcare system evolved continuously through this period. This pattern suggests that the January 2023 discontinuity in ambulance response times and the subsequent changes in health outcomes was not mechanically driven by a sudden alleviation of seasonal flu pressures.

5.4 Alternative Mechanism: Hospital Bed Availability and Capacity

The discharge funding introduced during the winter of 2022–2023 was intended to accelerate hospital patient discharge and relieve pressure on emergency departments. One potential channel through which this policy could reduce ambulance handover delays is by increasing the availability of hospital beds. If the funding rapidly freed hospital capacity, we would expect to observe an increase in the proportion of available beds around the same time as the decline in ambulance handover delays.

To examine this possibility, we use daily hospital bed availability data from the NHS England Urgent and Emergency Care Daily Situation Reports. These data report the availability of several types of beds across NHS Foundation Trusts, including adult General & Acute beds and adult critical care beds. We focus on adult beds, as these are most relevant for emergency admissions related to cardiac arrest and other severe conditions.

We estimate regression discontinuity in time (RDiT) specifications around the same cutoff used in the handover delay analysis, corresponding to the 5th January 2023. Table B.8 presents the estimated treatment effects for adult General & Acute bed availability and adult critical care bed availability. The effects are small and statistically insignificant across specifications, indicating no

clear discontinuity in bed availability at the policy cutoff. Figure B.21 presents the corresponding RDiT plots, which show concatenate trends in bed availability around the cutoff date.

These findings suggest that the improvement in ambulance performance documented in the main text was unlikely to have been driven by a sudden increase in the stock of available hospital beds. Instead, the policy likely improved the *flow of patients through the hospital system*, rather than largely increasing the number of empty beds at any given point in time.

This interpretation is consistent with how the discharge funding was implemented in practice. Much of the funding was used to expand temporary care placements, home care packages, and discharge coordination capacity. Such measures can accelerate patient throughput and reduce congestion in emergency departments without necessarily increasing measured bed availability at midnight, which is how bed availability is typically recorded in administrative statistics.

The absence of a discontinuity in bed availability combined with the sharp improvement of hospital discharge rates documented in Section 5.1 suggests that the primary operational effect of the policy was to improve patient flow through hospitals. By reducing the time ambulances spent waiting to transfer patients, hospitals effectively increased ambulance availability in the field, thereby shortening response times to new emergency calls.

5.5 Measuring Returns to the Discharge Fund

To gauge the economic efficiency of the £700 million Discharge Fund, we compare programmatic costs to the monetized value of the mortality reductions attributable to the policy. Because the funding improved patient flow across *all* emergency conditions—stroke, sepsis, trauma, and broader inpatient groups—estimating returns solely from the cardiac arrest channel provides a strict lower bound on total welfare value. We frame the exercise accordingly: the calculations below are a back-of-the-envelope accounting exercise designed to establish a floor.

Lives saved. Our DiDC estimate implies a 2.4 percentage point increase in 30-day survival for out-of-hospital cardiac arrest patients attributable to the policy discontinuity. Applying this estimate to the 26,113 cardiac arrest patients treated during the funded winter period (January–March 2023) yields approximately 627 additional survivors ($26,113 \times 0.024$).[11] The implied cost per life

[11] We restrict the treated population to January–March 2023, the period over which the Discharge Fund was operationally active, rather than extrapolating the local discontinuity estimate to a longer window. This is conservative: to the extent that decongestion effects persisted beyond March, the true number of beneficiaries is larger.

saved is:

$$\text{Cost per life saved} = \frac{£700{,}000{,}000}{627} \approx £1{,}116{,}507 \text{ per life.} \tag{4}$$

QALY-based valuation. We convert survival gains into Quality-Adjusted Life Years (QALYs) using the standard discounted utility formula:

$$\text{QALYs per survivor} = \sum_{t=1}^{T} \frac{\text{Utility}}{(1+r)^t}, \tag{5}$$

where we have several parameters to set up: (1) T stands for the life expectancy for survived cardiac arrest patients, we choose 10 years as a long-term study finds that about 64% of survivors are still alive after 10 years (Amacher *et al.*, 2022); (2) Utility stands for the measure of quality of life from 0 (death) to 1 (perfect health). Cardiac arrest survivors are considered being with lower utility score compared to general healthy population as they usually have other accompanying underlying diseases, and if there is delayed bystander CPR or other emergency treatments when a heart attack occurs, the sequelae of surviving patients will also affect the quality of life. The utility score is generally measured by self-reported EuroQol Health Questionnaire from the cardiac arrest survivors which could be integrated into a EuroQol index score. Existing medical literature show mixed results of this index score for cardiac arrest survivors (0.76 in Alm-Kruse *et al.* (2024); 0.82 in Smith *et al.* (2015); 0.9 in Yonis *et al.* (2023)), we choose 0.76 to calculate a conservative lower bound of health gains from the policy; (3) r stands for the discount rate, which is set as 3.5%, suggested by the NICE Guidelines. Thus, the QALYs per survivor is calculated as 6.32 QALYs. The total QALYs generated is 3962.6 QALYs, and the cost per QALY is £176,634 per QALY. These parameters are calculated as follows:

$$\text{Total QALYs gained} = 627 \times 6.32 = 3{,}962.6, \tag{6}$$

$$\text{Cost per QALY} = \frac{£700{,}000{,}000}{3{,}962.6} \approx £176{,}634. \tag{7}$$

At the NICE willingness-to-pay threshold of £20,000–£30,000 per QALY, the monetised health benefit from the cardiac arrest channel alone is:

$$\text{NICE-threshold return} = 3{,}962.6 \times £20{,}000 = £79.3 \text{ million.} \tag{8}$$

Table 4: Lower-Bound Returns to the £700m Discharge Fund: Cardiac Arrest Channel

Metric	Value
Total Programmatic Cost	£700,000,000
Treated Population (Jan–Mar 2023)	26,113
DiDC Survival Effect	2.4 pp
Estimated Lives Saved	627
Cost per Life Saved	£1,116,507
QALYs per Survivor	6.32
Total QALYs Gained	3,962.6
Cost per QALY	£176,634
NICE WTP Return (at £20k/QALY)	£79.3 Million

Why this lower bound is informative. The cost per QALY of £176,634 exceeds the standard NICE threshold, and the monetized cardiac-arrest benefit of £79.3 million accounts for roughly 11% of the programme's £700 million cost. However, cardiac arrest patients represent only a small fraction of total emergency admissions. During our sample period, the 26,113 cardiac arrests constituted approximately 0.6% of the 4,202,151 total Finished Admission Episodes.[12] The Discharge Fund's primary mechanism—freeing hospital beds and reducing A&E crowding—operated on *all* emergency and elective pathways. If the survival and morbidity gains among the remaining 99.5% of patients were even modestly positive, the aggregate return would be an order of magnitude larger, and the implied cost per QALY would fall correspondingly. In other words, the mortality benefit from this single, narrow subgroup is sufficient to offset more than one-tenth of the programme's total cost—before accounting for the policy's primary intended effect of relieving pressure on general acute care.

To place these implied returns in context, Table 5 benchmarks our estimates against existing work on the returns to health spending. Our implied cost per life saved (£1.1 million) and cost per QALY (£177k) are higher than the comparator estimates, as expected given that we capture only one narrow downstream channel of a system-wide intervention. Those cost-effectiveness calculations are intentionally conservative and should be treated as illustrative rather than definitive. We restrict the treated population to January–March 2023 and measure benefits only through the cardiac arrest channel. We do not estimate gains for other emergency conditions, for non-emergency hospital patients who may have benefited from reduced crowding, or for longer-run effects beyond the immediate winter period. Our calculations are therefore best understood as a lower bound that establishes the economic plausibility of large cross-stage returns, rather than a comprehensive welfare analysis.

[12] Data from Hospital Episode Statistics, NHS England(NHS England).

Table 5: Illustrative Benchmarks: Returns to Health Spending in Related Literature

Study	Intervention / Setting	Cost per Life Saved	Cost per QALY (or LY)
This paper	NHS Discharge Fund (cardiac arrest channel only)	£1,116,507	£176,634
Doyle *et al.* (2015)	*Higher-cost hospitals*	—	*$80,000*
Almond *et al.* (2010)	*Birth weight 1500g threshold*	*$527,083*	—
Cutler (2007)	*Revascularisation*	—	*$40,000*

Notes: "LY" denotes life-year. Our figures reflect a strict lower bound from a single downstream channel; the comparator studies estimate returns across the full treated population. Where studies report mortality effects without explicit cost-effectiveness metrics, the implied figures can be constructed from the reported mortality impact and programme costs. Cross-country and cross-period comparisons should be interpreted cautiously given differences in healthcare systems, price levels, and discount rates.

6 Discussion and Concluding Remarks

This paper set out to answer a simple question: when congestion is relieved at one stage of a multi-stage public service system, do the benefits propagate to other stages? Our evidence from the English emergency healthcare system suggests that they do, and that the spillovers are large enough to affect patient survival. Exploiting a £700 million discharge fund that targeted hospital patient discharge without directly intervening in ambulance services, we find that the policy increased return of spontaneous circulation for cardiac arrest patients by 3.8 percentage points and 30-day survival by 2.4 percentage points, operating through reduced ambulance handover delays and the release of vehicles for emergency dispatch.

The mechanism evidence points to patient throughput rather than bed capacity as the operative channel. The discharge fund did not produce a measurable increase in available hospital beds, yet it generated a sharp and sustained improvement in discharge rates, a reduction in ambulance handover delays, and ultimately better cardiac arrest outcomes. This pattern is consistent with a system in which the binding constraint was not the total number of beds but the speed at which patients moved through them. The practical implication is that policies aimed at unblocking patient flow through discharge coordination, step-down care commissioning, or social care capacity may be more effective per pound spent than policies that expand the physical stock of hospital capacity, at least in systems already operating near full occupancy.

The cross-stage structure of the spillover also matters for how we interpret cost-effectiveness. Any conventional evaluation of the discharge fund would naturally assess it in terms of hospital

throughput: how many additional patients were discharged, how much did length of stay fall, and at what cost per bed-day freed. Our results show that this evaluation framework, while not wrong, is incomplete. The cardiac arrest patients who survived because ambulances reached them faster would not appear in any hospital discharge metric. Their benefit occurred before they ever reached a hospital.

More broadly, our findings speak to the management of healthcare systems under acute pressure. The UK experience during the winter of 2022-2023 illustrated a pattern familiar to many publicly funded health systems: a surge in demand combined with reduced discharge capacity produced a cascading failure across the entire emergency pathway. The policy intervention was effective not because it added new resources to the system in a permanent sense, but because it relieved the specific bottleneck that was propagating congestion upstream. In interconnected systems operating under pressure, identifying and relieving the binding bottleneck may yield returns that far exceed what a localised assessment would predict.

Several limitations qualify these conclusions. Our identification strategy relies on the sharp timing of the discharge improvement around early January 2023. While the difference-in-discontinuities framework nets out seasonal patterns and the placebo cutoffs rule out recurring January effects, we cannot entirely exclude the possibility that other concurrent changes contributed to the observed improvements; we note, however, that influenza bed occupancy and hospital bed availability show no corresponding discontinuity at the cutoff, which narrows the set of plausible confounders. The regression discontinuity in time design estimates local effects at the cutoff, which may not generalise to other periods or to policy changes of different magnitudes as the winter of 2022-2023 represented an unusually severe congestion episode, and the estimated spillovers may be larger than what would be observed under more typical operating conditions. Our cost-effectiveness calculations are intentionally conservative and should be treated as illustrative: we restrict the treated population to January-March 2023, measure benefits only through the cardiac arrest channel, and do not estimate gains for other emergency conditions or longer-run effects. Finally, we observe the aggregate relationship between discharge improvement and cardiac arrest outcomes but cannot track individual patients through each stage of the congestion chain; the consistency of the evidence across multiple links provides reassurance, but patient-level linked data would allow a more definitive test of the causal pathway.

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Online Appendix – Not for Print

Congestion Spillovers in Multi-Stage Public Services: Evidence from England's Emergency Care System

Damian Clarke & Ye Yuan

A A stylized model of propagation of congestion in a multi-stage system

This appendix presents the formal model of multi-stage congestion referenced in Section 2.1. The model derives how demand shocks propagate from one stage to another when scarce resources are shared, and characterises the social planner's optimal allocation between two instruments, expanding capacity and improving patient flow, showing that flow-improving interventions yield dual welfare benefits.

We model emergency care as a multi-stage congested service system in which downstream hospital capacity constrains generate upstream delays. In this model, time is discrete, $t = 0, 1, 2, ...$, in each period, a random number of emergency patients A_t arrives and there is a fraction $\alpha \in (0, 1)$ of arrived patients require hospital admission. The admitted patients will occupy hospital beds until they are discharged. The number of occupied hospital beds at time t is B_t, which is less than or equal to the total staffed bed capacity K_t. Among those patients, there is a fraction of occupied beds δ_t released per period which is the discharge rate at time t. Bed occupancy evolves as:

$$B_{t+1} = B_t + \alpha A_t - \delta_t B_t. \tag{9}$$

We also define *hospital congestion* as utilization:

$$\rho_t = \frac{B_t}{K_t}, \tag{10}$$

which captures the extent to which downstream capacity is exhausted. When ρ_t is close to 1, the hospital is operating at or near full capacity, and which is less able to absorb incoming patients, generating ambulance handover delays:

$$H_t = h(\rho_t), \quad h'(\rho_t) > 0, \quad h''(\rho_t) \geq 0, \tag{11}$$

where H_t is the ambulance handover delay at time t, which is a function of hospital congestion ρ_t. The mapping from bed utilization to handover delay is likely to be convex because systems operating near full capacity have limited slack and are less able to absorb marginal increases in demand.

Ambulance delayed at hospitals become unavailable for responding to subsequent calls as they are occupied waiting to transfer current patient. Therefore, ambulances response time in period $t+1$ depends on handover delays in period t:

$$R_{t+1} = r(H_t), \quad r'(H_t) > 0, \tag{12}$$

which implies that higher hospital congestion leads to longer handover delays, which in turn leads to longer ambulance response times:

$$R_{t+1} = r(h(\rho_t)). \tag{13}$$

Above setting illustrates that the congestion in a multi-stage system can propagate dynamically: higher hospital congestion ρ_t leads to longer ambulance handover delays H_t, which in turn leads to longer ambulance response times R_{t+1} for

patients waiting in the community.

The planner intends to reduce congestion through two channels. First, increasing hospital capacity K_t can directly reduce hospital congestion ρ_t, let $x_t \geq 0$ be the additional investment in staffed bed capacity, then the new staffed capacity in the next period will be

$$K_{t+1} = K_t + x_t, \quad x_t \geq 0, \tag{14}$$

which leads to a reduction in hospital congestion ρ_{t+1}, and consequently reduces ambulance handover delays H_{t+1} and ambulance response times R_{t+2}. Second, increasing the discharge rate δ_t can also reduce hospital congestion ρ_t by accelerating patient discharge. Suppose $m_t \geq 0$ denotes the effort to improve discharge and patient flow:

$$\delta_{t+1} = \delta_t + m_t, \quad m_t \geq 0, \tag{15}$$

where the flow interventions include block-purchasing care home beds, buying more home care packages, and providing workforce support to increase staff hours and wages.

The social planner aims to minimize the discounted welfare loss from congestion and costs of interventions:

$$\min_{x_t, m_t} \sum_{t=0}^{\infty} \beta^t [L(R_t) + V(B_t) + C_K(x_t) + C_\delta(m_t)], \tag{16}$$

where $L(R_t)$ is the welfare loss from slow ambulance response times, which could be relevant for a range of health outcomes such as higher mortality, greater morbidity, and lower quality of life especially for patients with time-critical cases. To distinguish more sharply between capacity expansion and discharge-improving interventions, suppose the planner also faces a direct cost of high bed occupancy $V(B_t)$, denoted $V(B_t)$, with $V'(B_t) > 0$ and $V''(B_t) \geq 0$. This term captures the idea that a large inpatient stock generates management frictions and operational strain beyond its effect on utilization alone. $C_K(x_t)$ and $C_\delta(m_t)$ are convex cost functions of investing in additional staffed bed capacity and improving discharge rate, respectively. The planner's problem is to choose the optimal level of interventions that balances the benefits of reducing congestion against the costs of those interventions.

Suppose there is a random shock to demand for emergency care, which leads to a temporary increase in the number of emergency patients arriving at time t, $A_t = \overline{A} + \epsilon_t$.

Proposition A.1 (Demand shocks increase congestion)**.** *A positive demand shock $\epsilon_t > 0$ increases bed occupancy, hospital utilization, handover delay, and next-period ambulance response times.*

Proof. According to the law of motion of bed occupancy, $\frac{\partial B_{t+1}}{\partial A_t} = \alpha > 0$. Thus B_{t+1} increases with the demand shock. Since $\rho_{t+1} = \frac{B_{t+1}}{K_{t+1}}$, hospital bed occupancy rises as well. Because $h'(\rho_t) > 0$ and $r'(H_t) > 0$, this increases H_{t+1} and R_{t+2}. □

Proposition A.2 (Capacity expansion reduces congestion)**.** *An increase in capacity K_t reduces congestion and improves ambulance performance.*

Proof. $\frac{\partial \rho_t}{\partial K_t} = -\frac{B_t}{K_t^2} < 0$, thus $K_t \uparrow \Rightarrow \rho_t \downarrow \Rightarrow H_t \downarrow \Rightarrow R_{t+1} \downarrow$. □

Proposition A.3 (Flow improvement reduces congestion)**.** *An increase in discharge rate δ_t reduces future congestion and improves ambulance performance.*

Proof. $\frac{\partial B_{t+1}}{\partial \delta_t} = -B_t < 0$, thus $\delta_t \uparrow \Rightarrow B_{t+1} \downarrow \Rightarrow \rho_{t+1} \downarrow \Rightarrow H_{t+1} \downarrow \Rightarrow R_{t+2} \downarrow$. □

Proposition A.4 (Nonlinear amplification near capacity)**.** *Suppose $h'(\rho) > 0$ and $h''(\rho) > 0$, then for any two utilization levels $0 \le \rho_L < \rho_H \le 1$, and any increment $\Delta > 0$ such that $\rho_H + \Delta \le 1$,*

$$h(\rho_H + \Delta) - h(\rho_H) > h(\rho_L + \Delta) - h(\rho_L). \tag{17}$$

Thus, the same increase in hospital utilization generates a larger increase in handover delays when the system is already more congested.

Proof.

$$h(\rho + \Delta) - h(\rho) = \int_{\rho}^{\rho+\Delta} h'(u)\,du. \tag{18}$$

Since $h'(u)$ is increasing, the integral over the higher interval $[\rho_H, \rho_H + \Delta]$ is greater than the integral over the lower interval $[\rho_L, \rho_L + \Delta]$, hence

$$\int_{\rho_H}^{\rho_H+\Delta} h'(u)\,du > \int_{\rho_L}^{\rho_L+\Delta} h'(u)\,du, \tag{19}$$

which implies

$$h(\rho_H + \Delta) - h(\rho_H) > h(\rho_L + \Delta) - h(\rho_L). \tag{20}$$

□

The mapping from bed utilization to handover delay is likely to be convex because systems operating near full capacity have limited slack and are less able to absorb marginal increases in demand.

Corollary A.4.1 (Asymmetric gains from congestion relief)**.** *Under the conditions of Proposition A.4, a given reduction in congestion produces a larger decrease in handover delay when initial congestion is high than when it is low.*

Proof. Let $\Delta > 0$ be the magnitude of a reduction in congestion, then

$$h(\rho) - h(\rho - \Delta) = \int_{\rho-\Delta}^{\rho} h'(u)\,du. \tag{21}$$

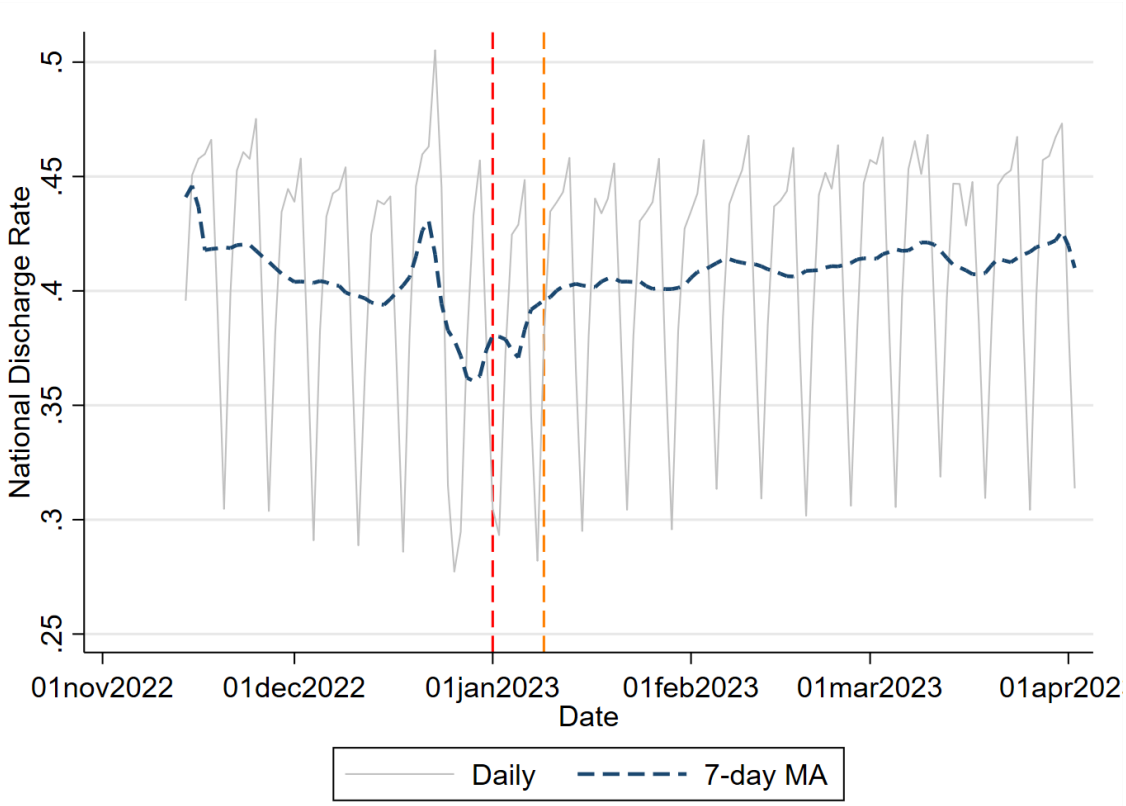
Since $h'(u)$ is increasing, the integral over the higher interval $[\rho_H - \Delta, \rho_H]$ is greater than the integral over the lower interval $[\rho_L - \Delta, \rho_L]$, the same reduction in hospital utilization generates a larger decrease in handover delays when the system is more congested. □

Proposition A.5 (The dual benefits of flow improvement)**.** *In the presence of an absolute volume penalty $V(B_t) > 0$, an increase in the discharge rate $m_t > 0$ yields a dual welfare benefit by reducing both future ambulance response times $L(R_{t+3})$ and absolute hospital volume costs $V(B_{t+2})$, while an increase in capacity K_t only reduces ambulance response times without directly reducing the hospital volume cost.*

Proof. Suppose $V'(B) > 0$, under the transition Equations (9), (14), and (15), a marginal increase in discharge effort m_t reduces the future bed occupancy and hence lowers the direct occupancy cost:

$$\frac{\partial V(B_{t+2})}{\partial m_t} = V'(B_{t+2})\frac{\partial B_{t+2}}{\partial m_t} = -V'(B_{t+2})B_{t+1} < 0. \tag{22}$$

Figure B.1: National Trends of Discharge Rate and 7-day Moving Average



By contrast, a marginal increase in capacity x_t does not directly affect next-period occupancy:

$$\frac{\partial B_{t+1}}{\partial x_t} = 0, \tag{23}$$

and therefore does not directly reduce the occupancy penalty through B_{t+1}. □

The model implies there is a steady state given above setting:

$$B^* = \frac{\alpha A}{\delta}, \quad \rho^* = \frac{\alpha A}{\delta K}. \tag{24}$$

In steady state, bed occupancy is determined by the ratio of demand to discharge rate, implying that hospital utilization is determined by the ratio of demand to discharge rate relative to capacity. This expression highlights two key channels through which the planner can reduce congestion: by increasing staffed bed capacity K or by improving discharge rates δ. Moreover, the model illustrates how shocks to demand can propagate through the system, leading to increased congestion and longer ambulance response times, and how targeted interventions can mitigate these effects.

B Appendix Figures and Tables

Figure B.2: Ambulance service operation

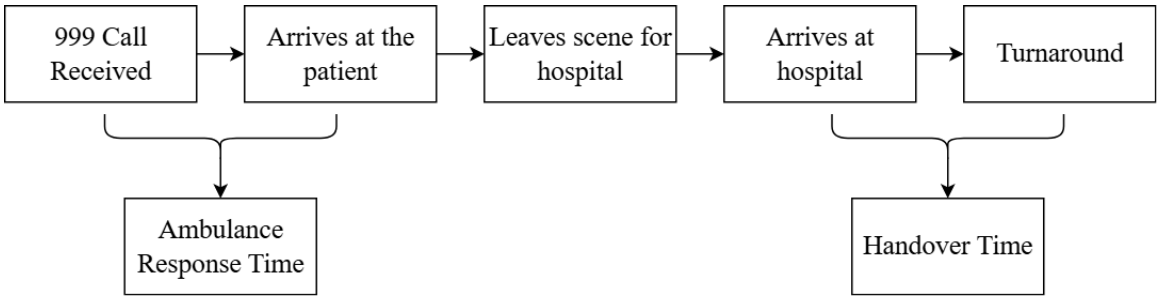


Figure B.3: Temporal Distribution of Adult Social Care Discharge Fund Disbursements

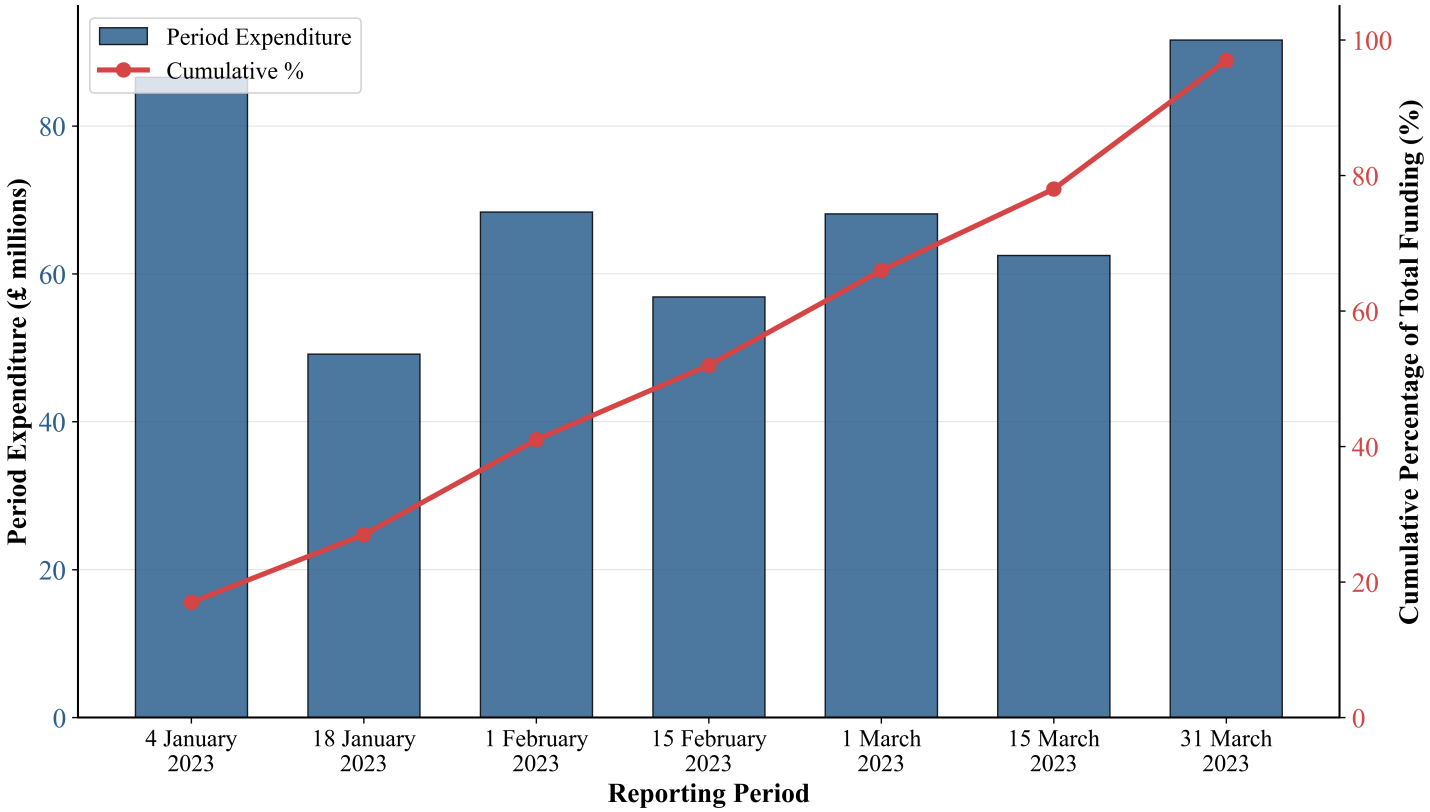


Figure B.4: Monthly Trends of Ambulance Data and Mortality Rate

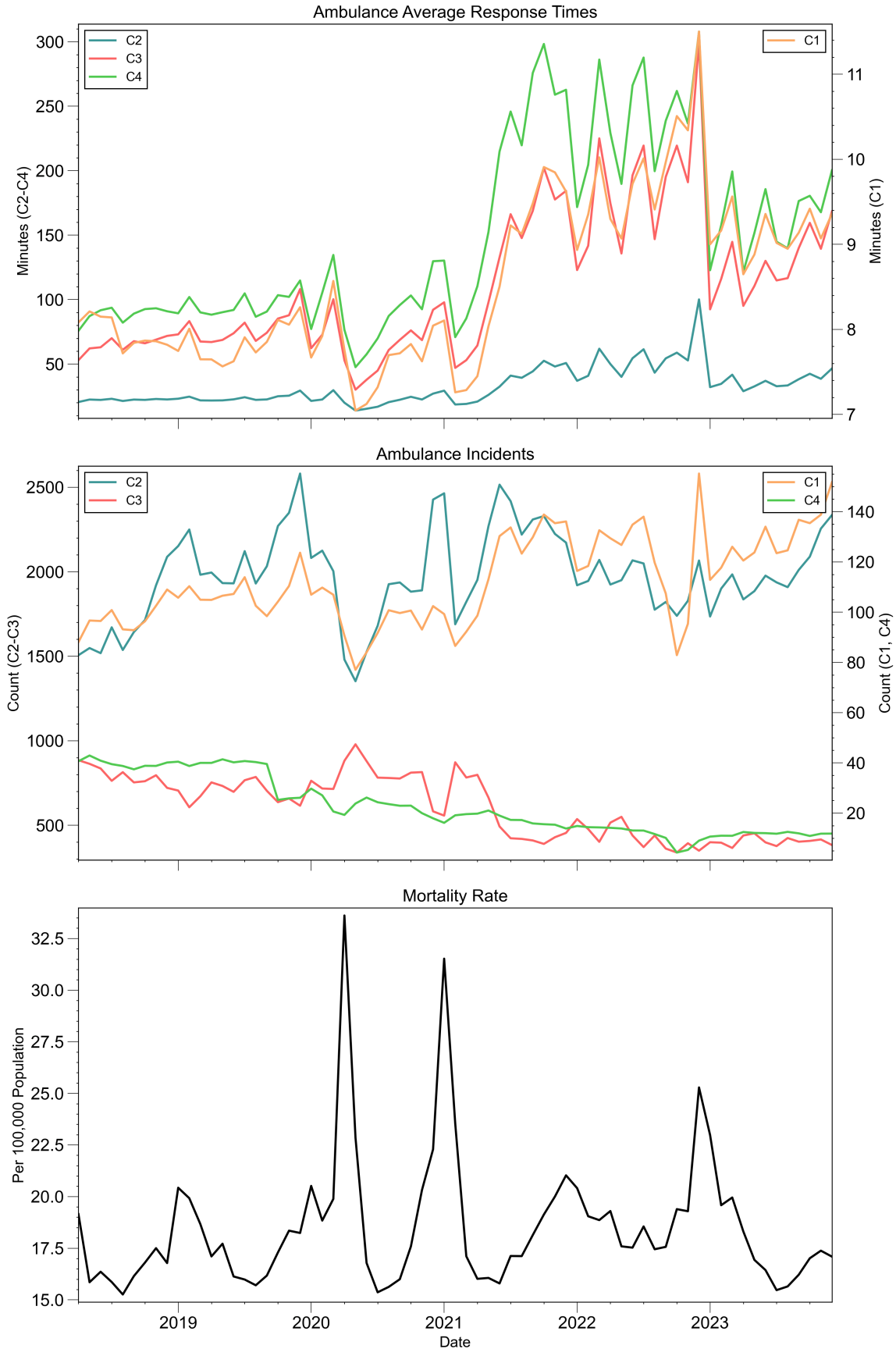


Figure B.5: Weekly Trends of Ambulance Data and Mortality Rate



Figure B.6: Ambulance Incidents and Response Times by Postcode Districts

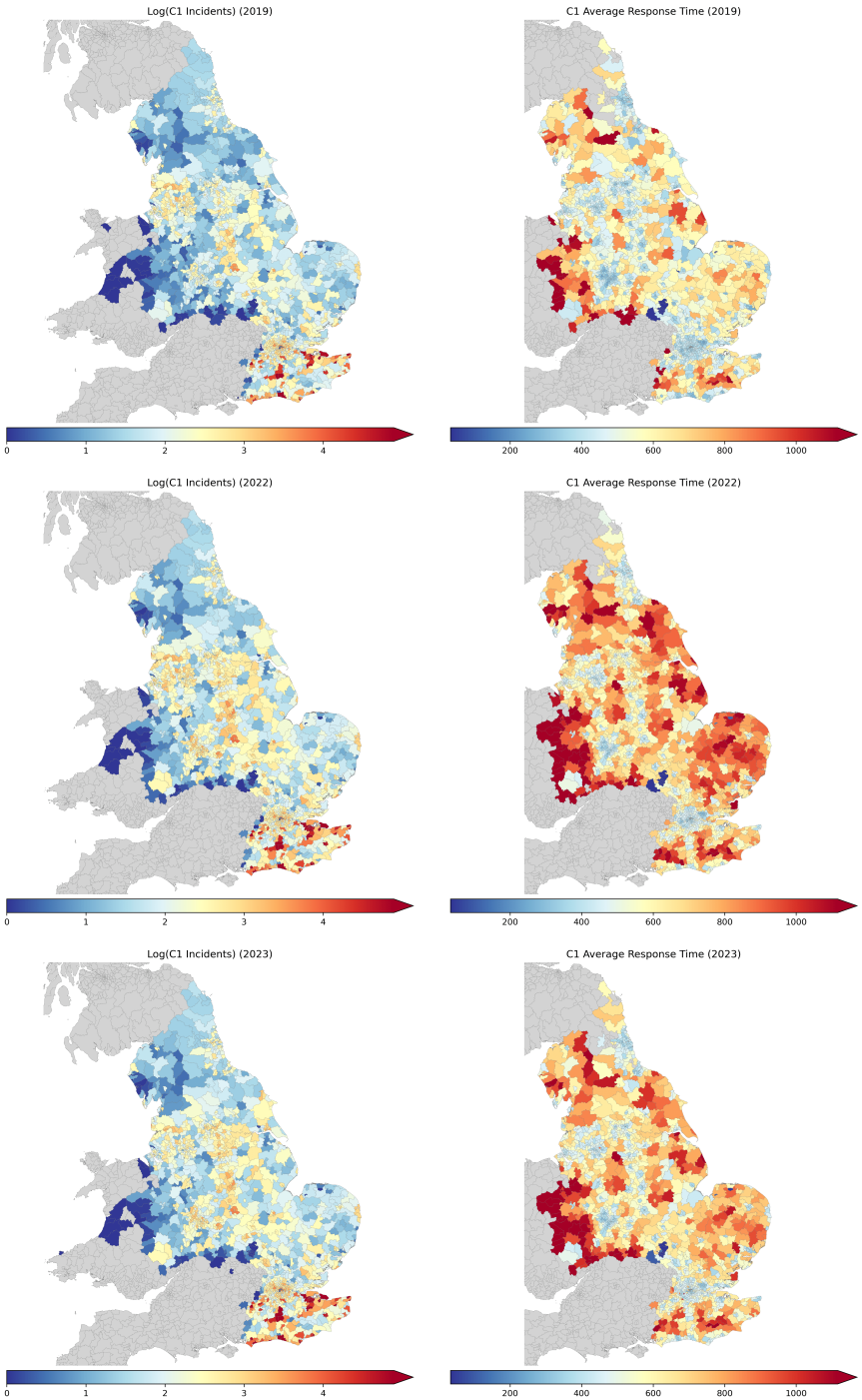


Figure B.7: Ambulance Data and Mortality Rate by Local Authorities

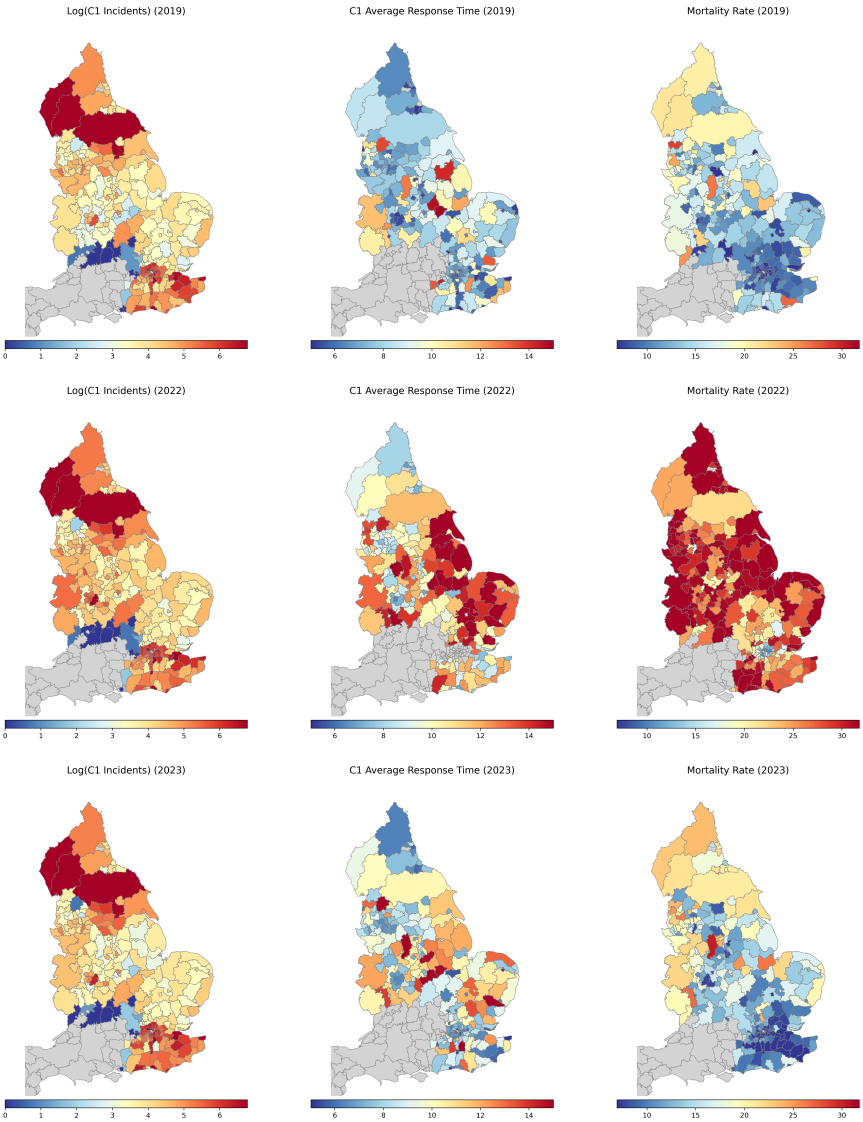
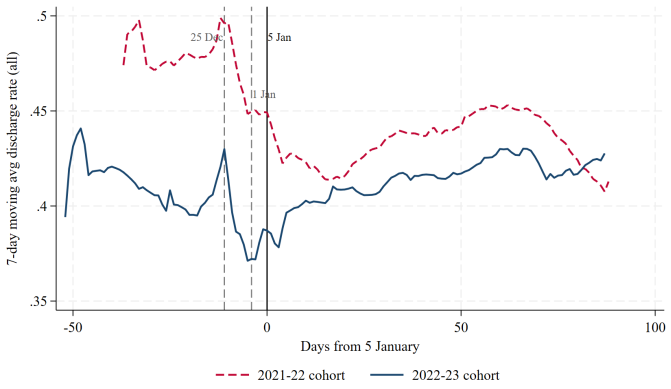
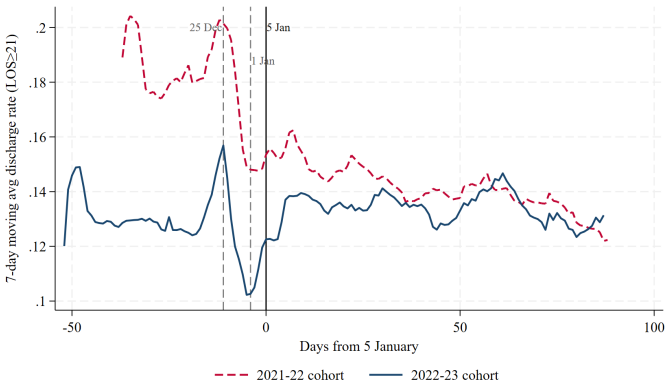


Figure B.8: Time Trends of 2021-2022 and 2022-2023 Hospital Discharge Rate by All Patients and Long Stay Patients

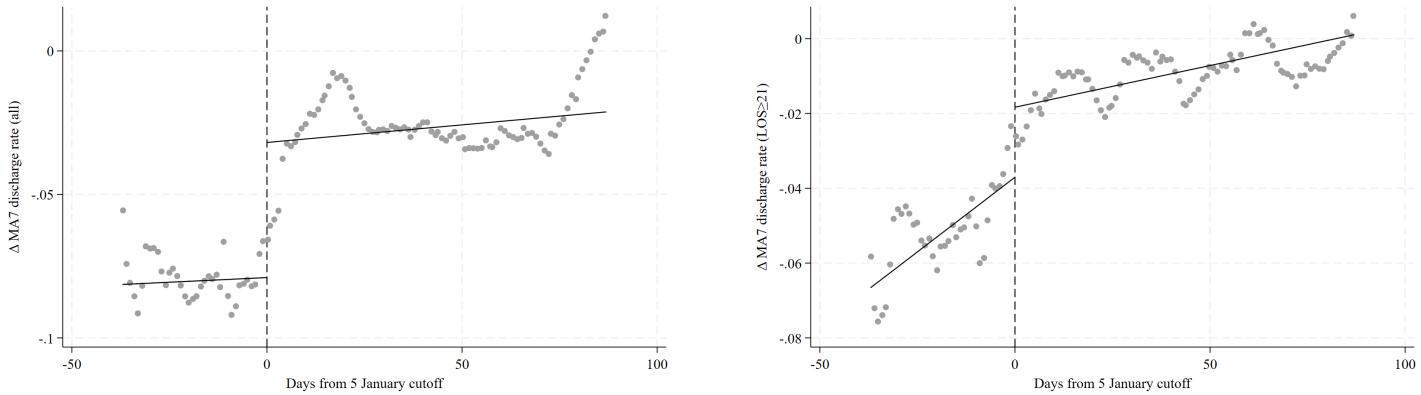


(a) All Patients



(b) Long Stay Patients

Figure B.9: Regression Discontinuity in Time: Year-over-Year Differenced Hospital Discharge Rate by All Patients and Long Stay Patients



(a) All Patients

(b) Long Stay Patients

Table B.1: RDD Results: Hospital Discharge Rate

	(1) Main RDD	(2) Placebo 2022	(3) Placebo 2024	(4) DiDC	(5) East England
Panel A: Discharge rate, all patients					
Treatment effect	0.028***	0.003	-0.011	0.029	0.056
Standard error	(0.007)	(0.014)	(0.008)	(0.030)	(0.129)
P-value	[0.0000]	[0.7269]	[0.6243]	[0.4530]	[0.8933]
Observations	4867	3218	1252	4772	528
Mean outcome	-0.013	-0.005	-0.018	-0.045	-0.064
Bandwidth (days)	20.49	13.67	5.96	20.15	21.21
Panel B: Discharge rate, length of stay ≥21 days					
Treatment effect	0.014	0.033***	-0.003	-0.004	-0.012
Standard error	(0.016)	(0.010)	(0.020)	(0.026)	(0.096)
P-value	[0.3889]	[0.0000]	[0.5645]	[0.7770]	[0.8766]
Observations	3431	1301	1707	4933	324
Mean outcome	-0.004	0.004	-0.005	-0.029	-0.033
Bandwidth (days)	14.33	5.17	7.25	21.04	13.90
Cutoff Year	2023	2022	2024	2023	2023
East of England Only	No	No	No	No	Yes

Notes: Conventional point estimate; (robust bias-corrected SE); [robust bias-corrected p-value]. Discharge rate = patients discharged / patients no longer meeting the criteria to reside, at the acute trust × day level, demeaned by trust × day of week within each winter. Col. (1): RDiT specification at the 5 January 2023 cutoff (winter 2022/23); col. (2): placebo cutoff at 5 January 2022 (winter 2021/22); col. (3): placebo cutoff at 5 January 2024 (winter 2023/24 acute discharge situation report, Type 1 acute trusts); col. (4): year-on-year differenced outcome (same trust, 364 days earlier, matching day of week) at the Jan 2023 cutoff; col. (5): East of England trusts only for specification (4). Weighted by the daily number of patients no longer meeting the criteria to reside. Running variable: days from 5 January of the cutoff year. Local linear estimator, triangular kernel, MSE-optimal bandwidth. Standard errors and p-values from Calonico *et al.* (2014) robust bias-corrected inference, clustered by hospital trust. *** p<0.01, ** p<0.05, * p<0.10.

Table B.2: RDD Results: Ambulance Handover Delays

	(1) Main RDD	(2) Placebo 2022	(3) Placebo 2024	(4) DiDC	(5) East England
Panel A: Share of arrivals with handover delay >30 minutes					
Treatment effect	-0.037	0.014	-0.064***	-0.015	-0.038
Standard error	(0.016)	(0.015)	(0.015)	(0.026)	(0.122)
P-value	[0.1079]	[0.1701]	[0.0005]	[0.7929]	[0.7289]
Observations	2660	1629	1389	2564	362
Mean outcome	0.067	0.012	0.001	0.149	0.281
Bandwidth (days)	10.28	6.39	5.96	10.96	14.23
Panel B: Share of arrivals with handover delay >60 minutes					
Treatment effect	-0.054***	0.012	-0.092***	-0.013	-0.085
Standard error	(0.015)	(0.011)	(0.011)	(0.023)	(0.102)
P-value	[0.0025]	[0.1607]	[0.0000]	[0.5085]	[0.4081]
Observations	3169	1629	2649	2564	312
Mean outcome	0.046	0.008	0.007	0.107	0.216
Bandwidth (days)	12.12	6.14	10.27	10.12	12.77
Panel C: Hours lost to handover delays per arrival					
Treatment effect	-0.282***	—	-0.347***	—	-0.369*
Standard error	(0.080)		(0.037)		(0.190)
P-value	[0.0020]		[0.0000]		[0.0828]
Observations	3677	—	3149	—	525
Mean outcome	0.145	—	-0.007	—	0.209
Bandwidth (days)	14.33	—	12.77	—	19.98
Cutoff Year	2023	2022	2024	2023	2023
East of England Only	No	No	No	No	Yes

Notes: Conventional point estimate; (robust bias-corrected SE); [robust bias-corrected p-value]. Outcomes are daily handover delay measures at the trust × day level, demeaned by trust × day of week within each winter. Col. (1): RDiT specification at the 5 January 2023 cutoff (winter 2022/23 UEC daily situation reports); col. (2): placebo cutoff at 5 January 2022 (winter 2021/22 UEC daily situation reports; the hours-lost metric was not collected in 2021/22, so Panel C is not estimable); col. (3): placebo cutoff at 5 January 2024 (winter 2023/24 NHS England ambulance handovers collection, the successor to the sitrep series: shares are delays >30 and >60 minutes over handovers with known handover time, and Panel C uses handover hours beyond 30 minutes per handover); col. (4): year-on-year differenced outcome (same trust, 364 days earlier, matching day of week) at the Jan 2023 cutoff (Panel C not estimable); col. (5): East of England trusts only for specification (4) (Panel C: specification (1)). Weighted by daily ambulance arrivals (col. 3: handovers with known time). Running variable: days from 5 January of the cutoff year. Local linear estimator, triangular kernel, MSE-optimal bandwidth. Standard errors and p-values from Calonico *et al.* (2014) robust bias-corrected inference, clustered by hospital trust. *** p<0.01, ** p<0.05, * p<0.10.

Figure B.10: The Discontinuity of Ambulance Response Times, p=2

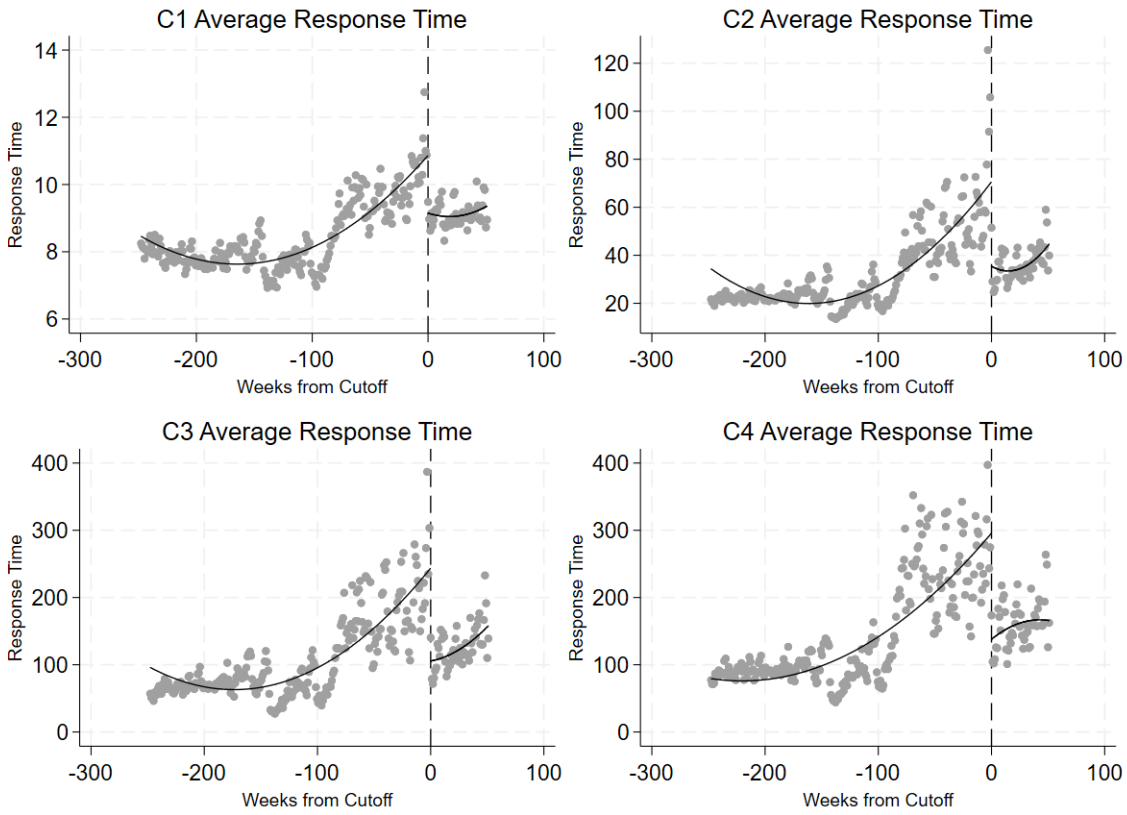
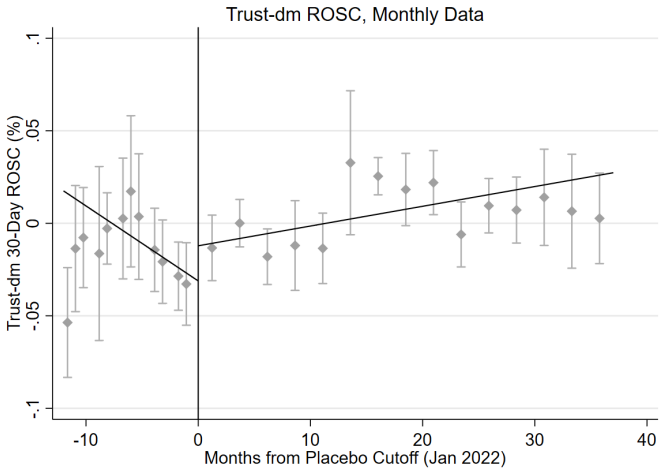
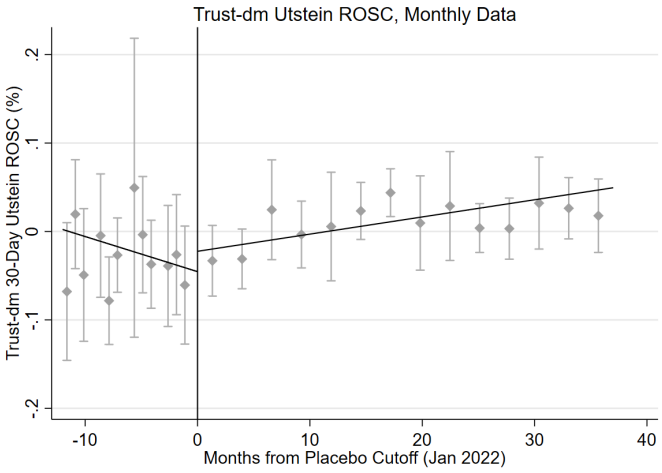


Figure B.11: Regression Discontinuity in Time: ROSC Rate at the placebo time cutoff January 2022



(a) ROSC Rate



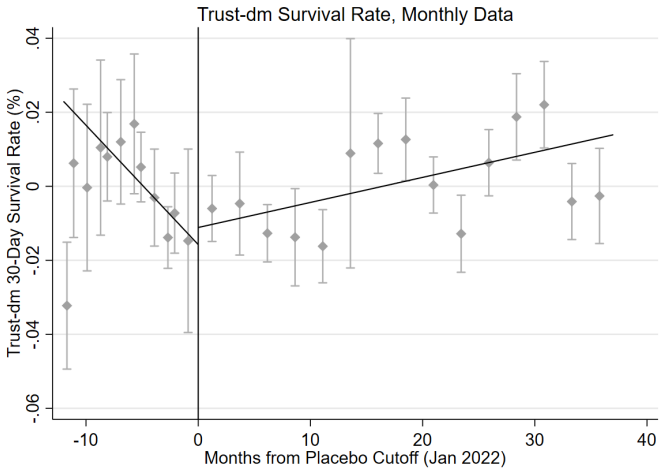
(b) ROSC Rate for Utstein Patients

Table B.3: RDD Results: Ambulance Response Times

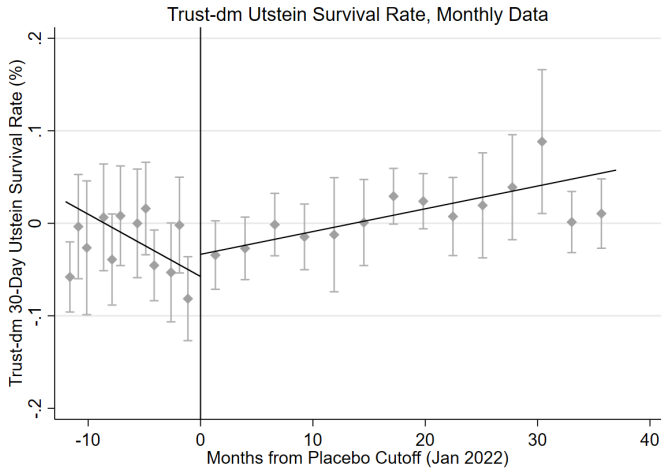
	(1) Main RDD	(2) Placebo 2022	(3) Placebo 2024	(4) DiDC	(5) East England
Panel A: C1 average response time (minutes)					
Treatment effect	-2.50***	-0.52***	—	-2.08***	-1.38***
Standard error	(0.13)	(0.08)		(0.17)	(0.26)
P-value	[0.0000]	[0.0000]		[0.0000]	[0.0000]
Observations	7655	11974	—	6612	1040
Mean outcome	1.22	0.90	—	0.19	-0.51
Bandwidth (weeks)	15.55	23.55	—	13.63	11.58
Panel B: C2 average response time (minutes)					
Treatment effect	-29.33***	-7.79***	—	-25.73***	-46.80***
Standard error	(1.86)	(0.73)		(1.72)	(4.43)
P-value	[0.0000]	[0.0000]		[0.0000]	[0.0000]
Observations	3827	17313	—	3320	788
Mean outcome	18.17	13.07	—	7.67	9.50
Bandwidth (weeks)	7.15	33.21	—	6.21	8.58
Panel C: C3 average response time (minutes)					
Treatment effect	-159.88***	-35.21***	—	-137.40***	-167.21***
Standard error	(7.59)	(4.76)		(10.38)	(16.47)
P-value	[0.0000]	[0.0000]		[0.0000]	[0.0000]
Observations	3803	6932	—	2818	687
Mean outcome	55.41	65.87	—	11.96	14.34
Bandwidth (weeks)	7.51	13.75	—	5.39	7.62
Panel D: C4 average response time (minutes)					
Treatment effect	-144.62***	-32.74***	—	-101.63***	-118.37**
Standard error	(17.44)	(9.43)		(23.44)	(72.88)
P-value	[0.0000]	[0.0090]		[0.0000]	[0.0473]
Observations	4908	7334	—	3824	730
Mean outcome	56.88	89.88	—	-45.52	3.26
Bandwidth (weeks)	13.05	20.80	—	13.35	21.83
Cutoff Year	2023	2022	2024	2023	2023
East of England Only	No	No	No	No	Yes

Notes: Conventional point estimate; (robust bias-corrected SE); [robust bias-corrected p-value]. Outcomes are local-authority-demeaned weekly average response times in minutes, winsorized at the 1st and 99th percentiles. Col. (1): RDiT specification at Jan 2023 cutoff; col. (2): placebo cutoff at Jan 2022; col. (3) is not estimable because the response time data end in December 2023; col. (4): year-on-year differenced outcome (same local authority and week, previous year) at Jan 2023 cutoff; col. (5): East of England local authorities only for specification (4). Local authority × week panel with category-specific incident-count weights. Running variable: weeks from January 2023 (col. 2: weeks from January 2022). Local linear estimator, triangular kernel, MSE-optimal bandwidth. Standard errors and p-values from Calonico *et al.* (2014) robust bias-corrected inference, clustered by local authority. *** p<0.01, ** p<0.05, * p<0.10.

Figure B.12: Regression Discontinuity in Time: Survival Rate at 30 Days at the placebo time cutoff January 2022

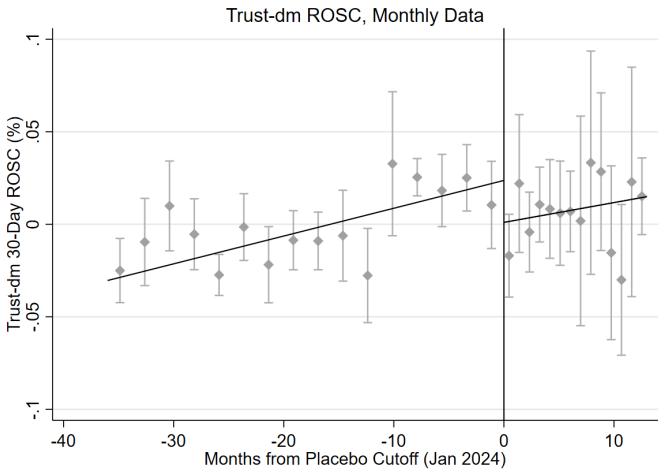


(a) Survival Rate at 30 Days

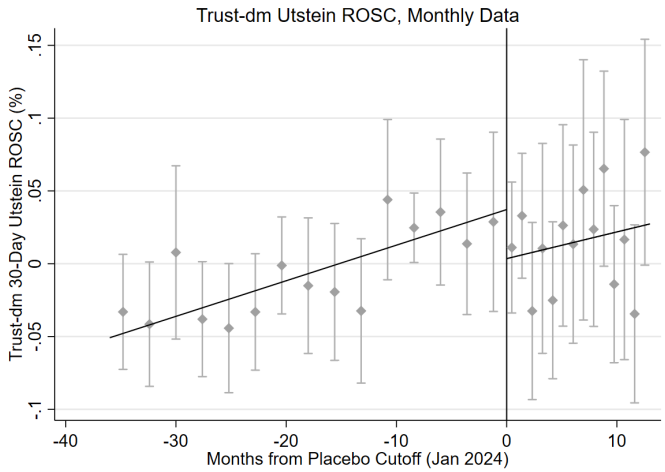


(b) Survival Rate at 30 Days for Utstein Patients

Figure B.13: Regression Discontinuity in Time: ROSC Rate at the placebo time cutoff January 2024

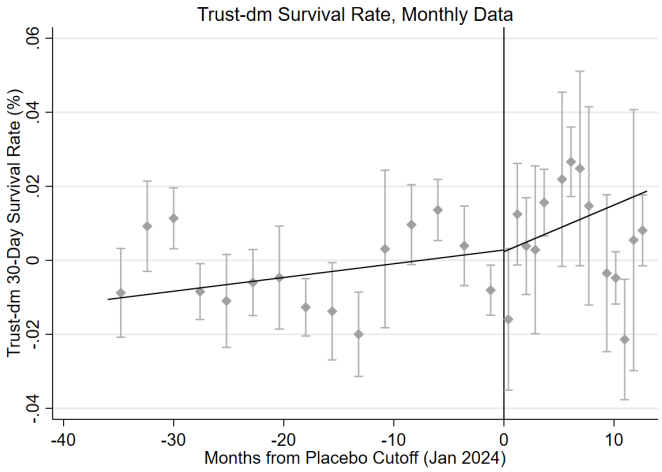


(a) ROSC Rate

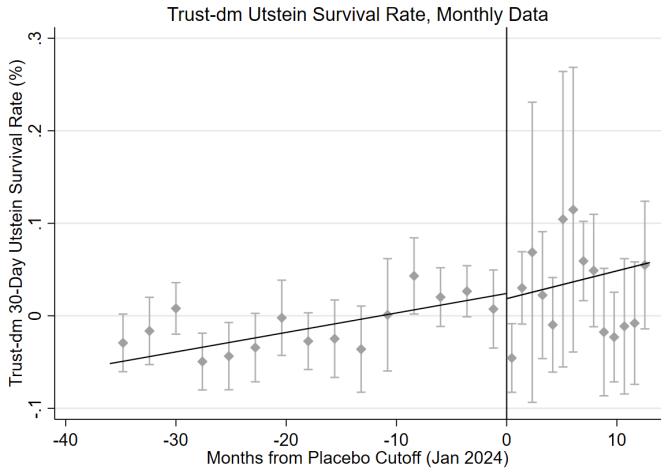


(b) ROSC Rate for Utstein Patients

Figure B.14: Regression Discontinuity in Time: Survival Rate at 30 Days at the placebo time cutoff January 2024

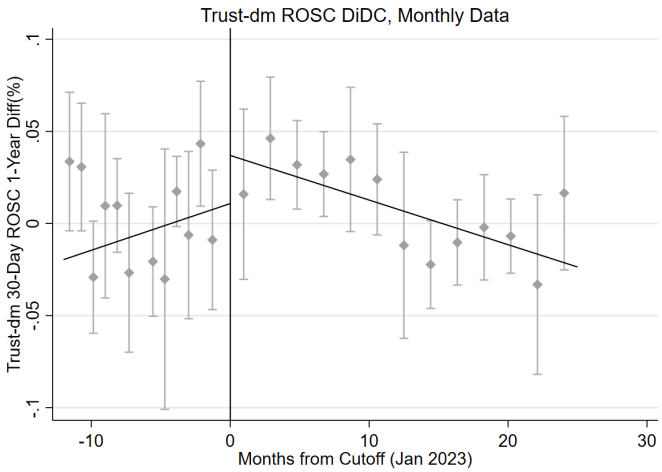


(a) Survival Rate at 30 Days

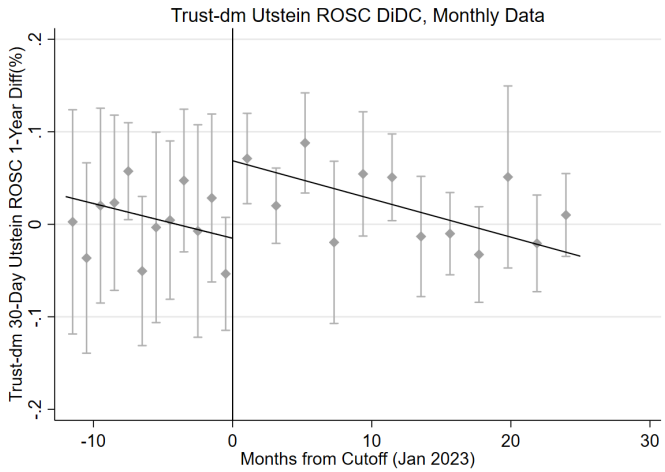


(b) Survival Rate at 30 Days for Utstein Patients

Figure B.15: Regression Discontinuity in Time: Year-over-Year Differenced ROSC Rate (DiDC)

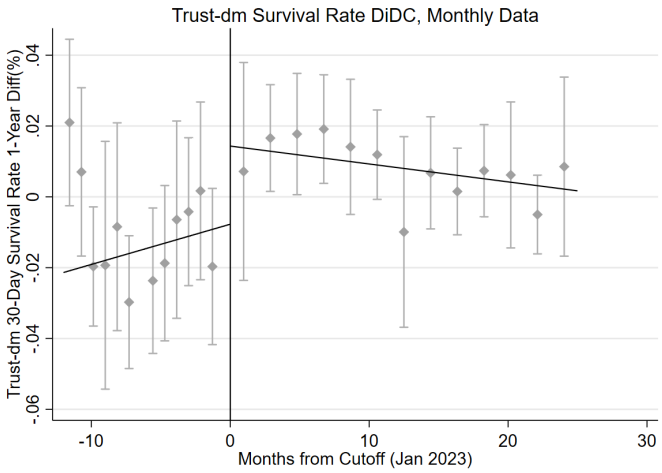


(a) ROSC Rate

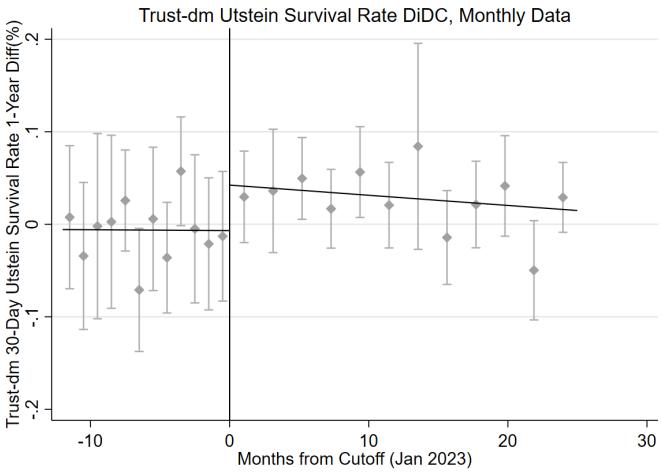


(b) ROSC Rate for Utstein Patients

Figure B.16: Regression Discontinuity in Time: Year-over-Year Differenced Survival Rate at 30 Days (DiDC)



(a) Survival Rate at 30 Days



(b) Survival Rate at 30 Days for Utstein Patients

Table B.4: Sensitivity Analysis: ROSC Outcomes (DiDC)

ROSC Rate			Utstein ROSC		
Parameter	Coefficient	S.D.	Parameter	Coefficient	S.D.
Panel A: Bandwidth sensitivity (local linear, $p = 1$)					
—(50%)	—	—	—(50%)	—	—
h= 2.07(60%)	0.0659	0.075	—(60%)	—	—
h= 2.41(70%)	0.0627	0.051	h= 2.19(70%)	0.2137***	0.037
h= 2.76(80%)	0.0615**	0.023	h= 2.51(80%)	0.2123***	0.045
h= 3.10(90%)	0.0500***	0.043	h= 2.82(90%)	0.2116***	0.031
h= 3.45(100%)	**0.0377*** **	**0.043**	**h= 3.13(100%)**	**0.1929****	**0.132**
h= 3.79(110%)	0.0334***	0.043	h= 3.45(110%)	0.1784**	0.132
h= 4.14(120%)	0.0299**	0.037	h= 3.76(120%)	0.1727**	0.132
h= 4.48(130%)	0.0274**	0.035	h= 4.07(130%)	0.1669***	0.091
h= 4.83(140%)	0.0261*	0.035	h= 4.39(140%)	0.1590***	0.070
h= 5.17(150%)	0.0232*	0.031	h= 4.70(150%)	0.1549***	0.068
Panel B: Polynomial order sensitivity (MSE-optimal bandwidth)					
p=1	**0.0377***	**0.031**	**p=1**	**0.1929*****	**0.070**
p=2	0.0697*	0.048	p=2	0.2062***	0.070
p=3	0.1193**	0.068	p=3	0.2642**	0.110

Notes: Dependent variables are trust-demeaned ROSC rates. Panel A tests sensitivity to bandwidth choice using linear polynomial ($p = 1$); bold row: MSE-optimal (100%). Bandwidth values differ between outcomes as each uses multiples of its own optimal bandwidth. Panel B tests sensitivity to polynomial order using MSE-optimal bandwidth for each outcome; bold row: baseline ($p = 1$). All specifications use triangular kernel, Jan 2023 cutoff, unbalanced panel (11 trusts), `r0n` weights, SE clustered by trust. Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.5: Sensitivity Analysis: 30-Day Survival Outcomes (DiDC)

30-Day Survival			Utstein Survival		
Parameter	Coefficient	S.D.	Parameter	Coefficient	S.D.
Panel A: Bandwidth sensitivity (local linear, $p = 1$)					
h= 2.08(50%)	0.0284	0.037	h= 2.87(50%)	0.0505	0.064
h= 2.50(60%)	0.0265***	0.007	h= 3.44(60%)	0.0544	0.084
—(70%)	—	—	h= 4.01(70%)	0.0553	0.081
h= 3.33(80%)	0.0252	0.021	h= 4.59(80%)	0.0633	0.083
h= 3.74(90%)	0.0247	0.021	h= 5.16(90%)	0.0614	0.083
h= 4.16(100%)	**0.0237****	**0.012**	**h= 5.73(100%)**	**0.0544**	**0.083**
h= 4.58(110%)	0.0228***	0.011	h= 6.31(110%)	0.0512	0.075
h= 4.99(120%)	0.0223***	0.011	h= 6.88(120%)	0.0500	0.072
h= 5.41(130%)	0.0189***	0.012	h= 7.45(130%)	0.0450	0.067
h= 5.82(140%)	0.0172***	0.012	h= 8.03(140%)	0.0417	0.065
h= 6.24(150%)	0.0152***	0.012	h= 8.60(150%)	0.0428	0.062
Panel B: Polynomial order sensitivity (MSE-optimal bandwidth)					
p=1	**0.0237*****	**0.012**	**p=1**	**0.0544**	**0.066**
p=2	0.0344**	0.016	p=2	0.0256	0.097
p=3	0.0124	0.075	p=3	0.0555	0.224

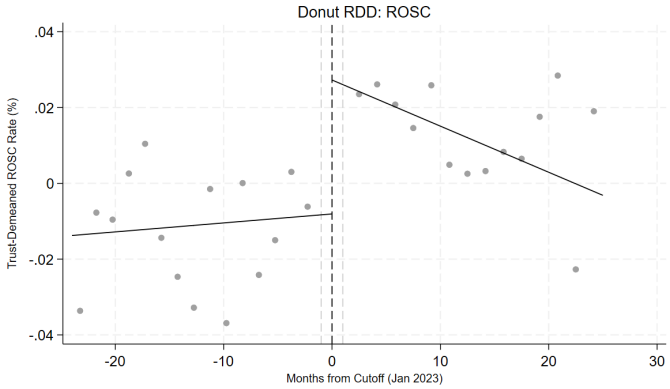
Notes: Dependent variables are trust-demeaned 30-day survival rates. Panel A tests sensitivity to bandwidth choice using linear polynomial ($p = 1$); bold row: MSE-optimal (100%). Bandwidth values differ between outcomes as each uses multiples of its own optimal bandwidth. Panel B tests sensitivity to polynomial order using MSE-optimal bandwidth for each outcome; bold row: baseline ($p = 1$). All specifications use triangular kernel, Jan 2023 cutoff, unbalanced panel (11 trusts), `r0n` weights, SE clustered by trust. Significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.6: Robustness: Panel and Weighting (DiDC, Jan 2023)

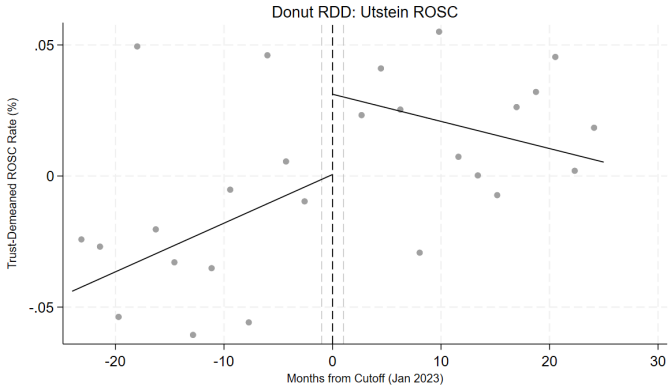
		ROSC	ROSC (Utstein)	Survival	Survival(Utstein)
(a) Balanced panel without weights	Coefficient	0.035*	0.227**	0.022***	0.066
		(0.030)	(0.107)	(0.012)	(0.065)
		[0.0666]	[0.0150]	[0.0076]	[0.1804]
	Bandwidth	3.25	2.75	3.76	4.76
	Observations	70	50	70	90
(b) Inbalanced panel without weights	Coefficient	0.066	0.232**	0.041	0.071
		(0.060)	(0.105)	(0.034)	(0.079)
		[0.1261]	[0.0129]	[0.1079]	[0.2841]
	Bandwidth	2.76	2.92	3.05	4.09
	Observations	55	52	76	93

Notes: DiDC (difference-in-discontinuities) at Jan 2023 cutoff. Conventional point estimate; (robust bias-corrected SE); [robust bias-corrected p-value]. Row (a): balanced panel excluding Isle of Wight (R1F), no weights. Row (b): full unbalanced panel including R1F, no weights. Main specification uses unbalanced panel with `r0n` weights (Tables 1–2). SE clustered by ambulance trust. *** p<0.01, ** p<0.05, * p<0.10.

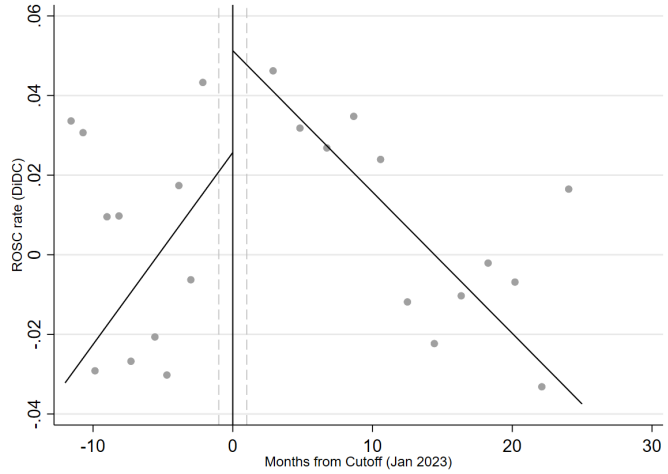
Figure B.17: ± 1 month Donut: ROSC Rate



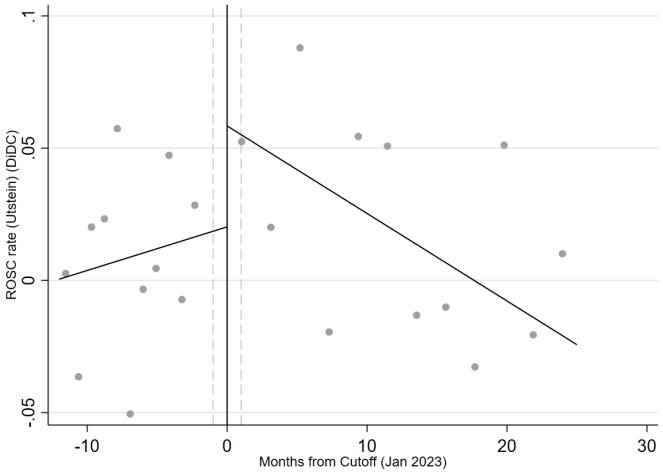
(a) ROSC Rate: RD Estimate



(b) ROSC Rate for Utstein Patients: RD Estimate



(c) ROSC Rate: DiDC Estimate



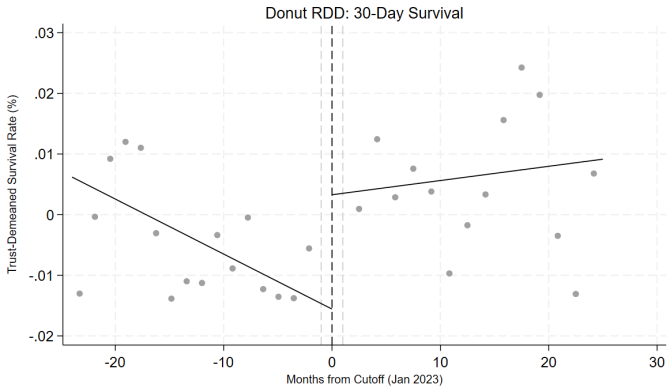
(d) ROSC Rate for Utstein Patients: DiDC Estimate

Table B.7: Robustness: DiDC with SE Clustered by Calendar Month or Trust × Time

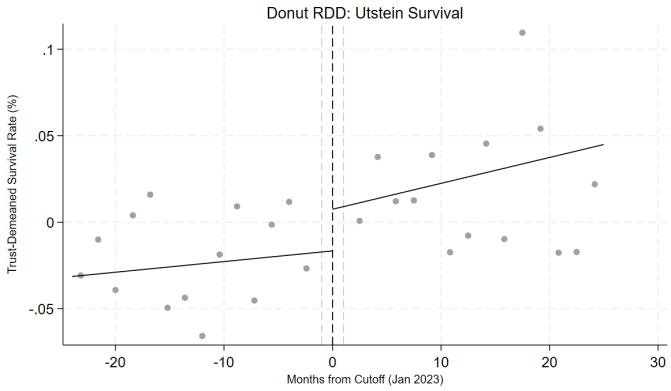
		ROSC	ROSC (Utstein)	Survival	Survival(Utstein)
(a) SE Clustered by Calendar Month	Coefficient	0.039***	0.176***	0.024***	0.055***
		(0.021)	(0.025)	(0.007)	(0.014)
		[0.0053]	[0.0000]	[0.0000]	[0.0000]
	Bandwidth	3.37	3.56	4.18	3.97
	Observations	76	73	97	73
	Mean DiDC	0.021	0.028	0.002	0.012
(b) SE Clustered by Trust × Time	Coefficient	0.051*	0.211**	0.025**	0.065
		(0.036)	(0.098)	(0.014)	(0.066)
		[0.0589]	[0.0134]	[0.0119]	[0.2274]
	Bandwidth	3.09	2.90	3.61	4.83
	Observations	76	52	76	93
	Mean DiDC	0.021	0.039	0.003	0.020

Notes: DiDC (difference-in-discontinuities) at Jan 2023 cutoff. Conventional point estimate; (robust bias-corrected SE); [robust bias-corrected p-value]. Panel (a) is identical to main specification except SE are clustered by *calendar month*. Panel (b) reports coefficients with SE clustered by trust× month (= HC1 robust SE since each trust-month is the observational unit). Both panels use unbalanced panel dataset (11 trusts)and number of cardiac arrest weights. Bandwidth in months. *** p<0.01, ** p<0.05, * p<0.10.

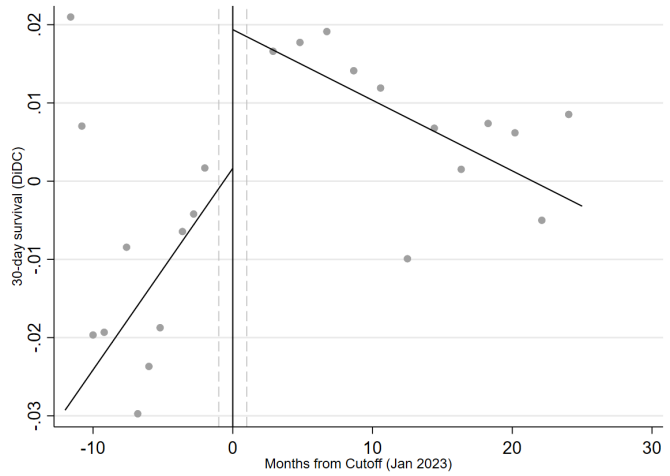
Figure B.18: ± 1 month Donut: Survival Rate at 30 Days



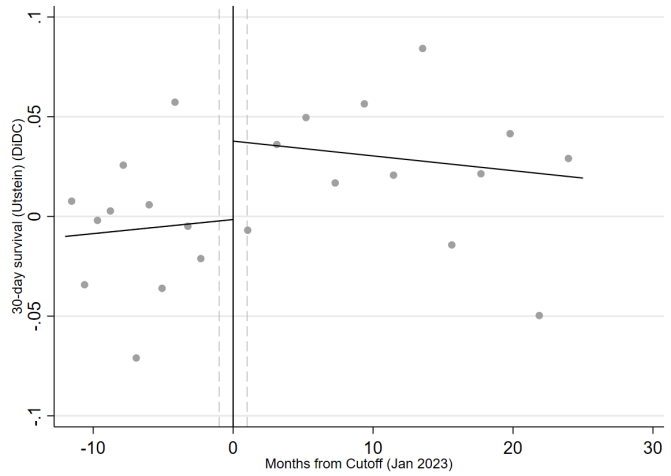
(a) Survival Rate at 30 Days: RD Estimate



(b) Survival Rate at 30 Days for Utstein Patients: RD Estimate



(c) Survival Rate at 30 Days: DiDC Estimate



(d) Survival Rate at 30 Days for Utstein Patients: DiDC Estimate

Table B.8: RDiT Estimates: Hospital Bed Availability

Bed Type	Coefficient	Std. Error	P-value	Bandwidth
Adult Critical Care Availability Rate	-0.0085	(0.0169)	0.617	20.18
Adult General & Acute Availability Rate	0.0012	(0.0059)	0.845	18.18

Notes: The table reports regression discontinuity in time (RDiT) estimates of the effect of the 5th January 2023 cutoff on hospital bed availability rates across NHS Foundation Trusts. All specifications use a triangular kernel with linear polynomial and MSE-optimal bandwidth selection. Standard errors are clustered at the trust level. Significance levels: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Figure B.19: Number of Cardiac Arrests by Trusts

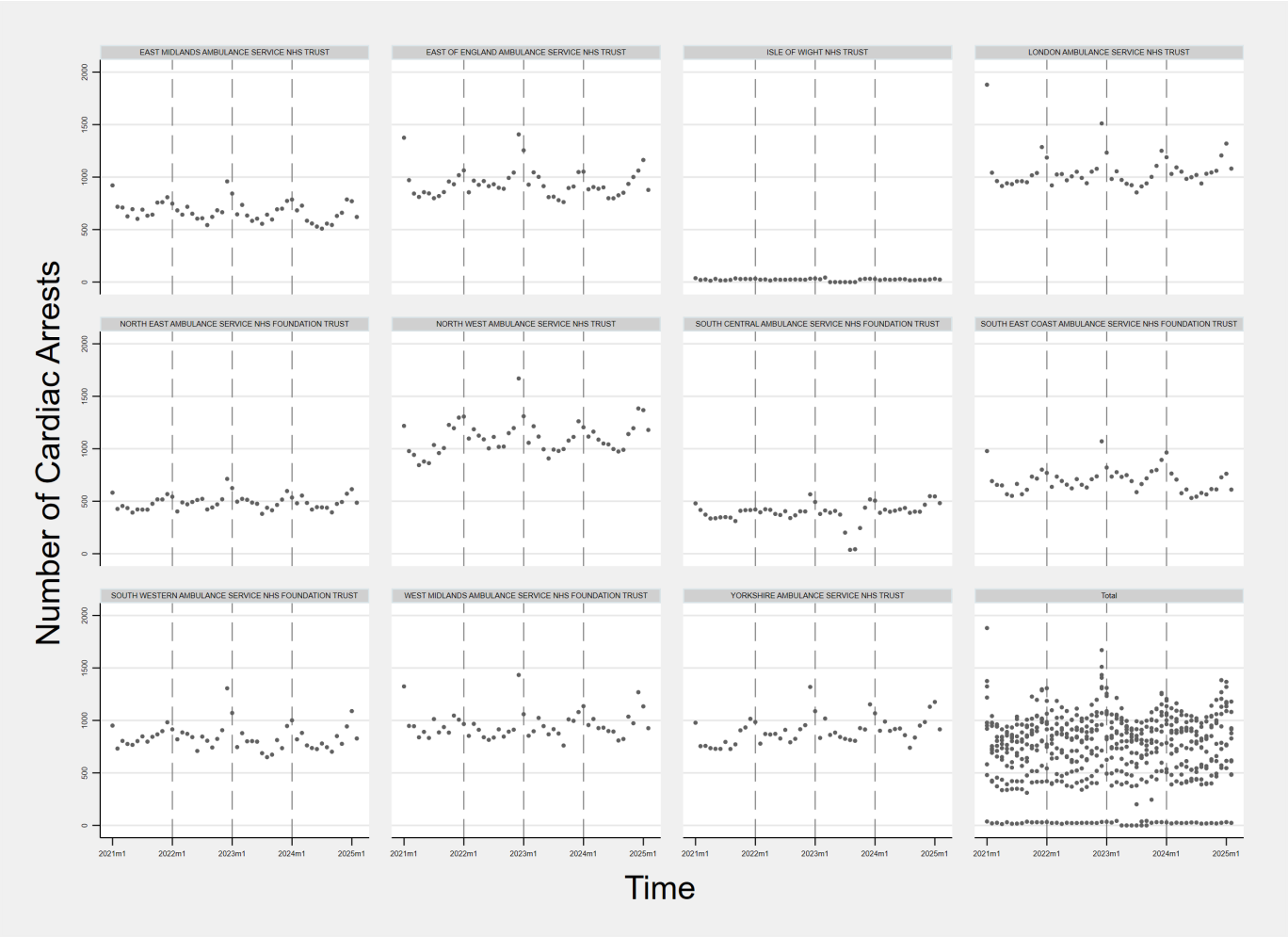
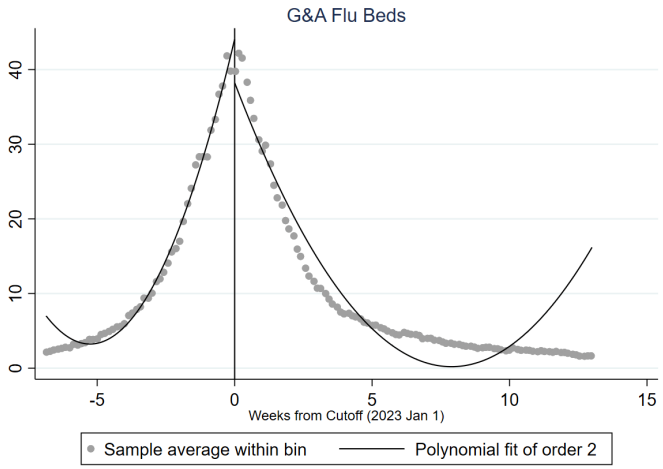
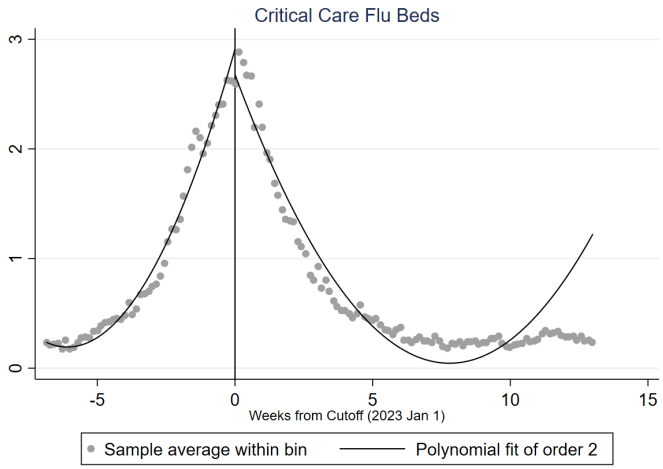


Figure B.20: Regression discontinuity plots of beds occupied by patients with laboratory-confirmed influenza

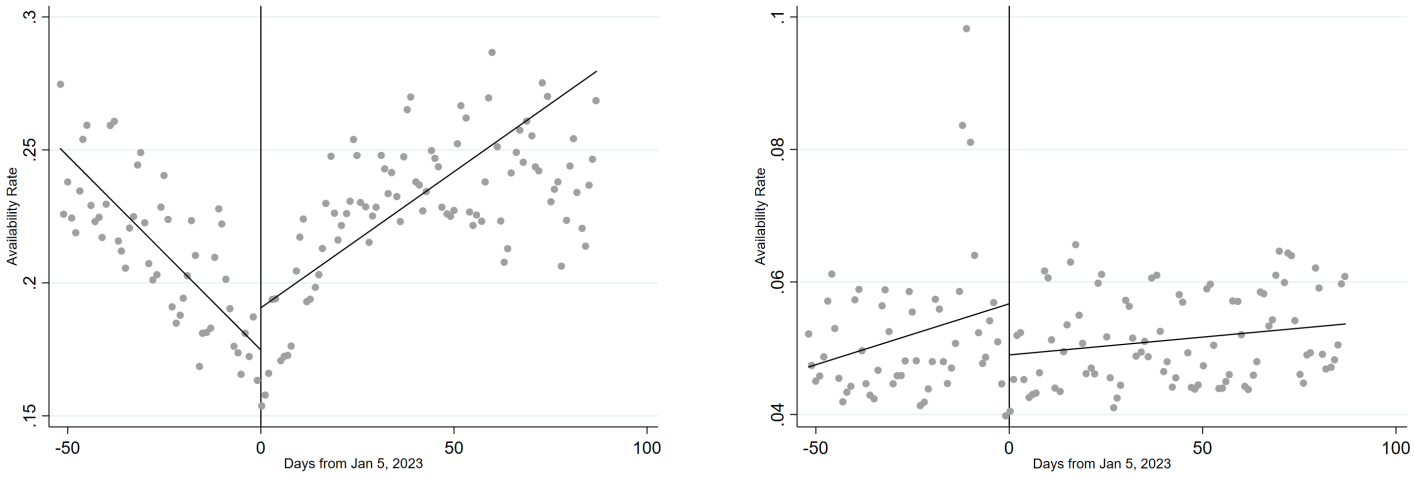


(a) General and Acute (G&A) Beds



(b) Critical Care (CC) Beds

Figure B.21: Regression Discontinuity in Time: Adult Hospital Bed Availability Rate



(a) Adult Critical Care Beds

(b) Adult G&A Beds

Figure B.22: Share of ambulance arrivals experiencing handover delays exceeding 30 minutes

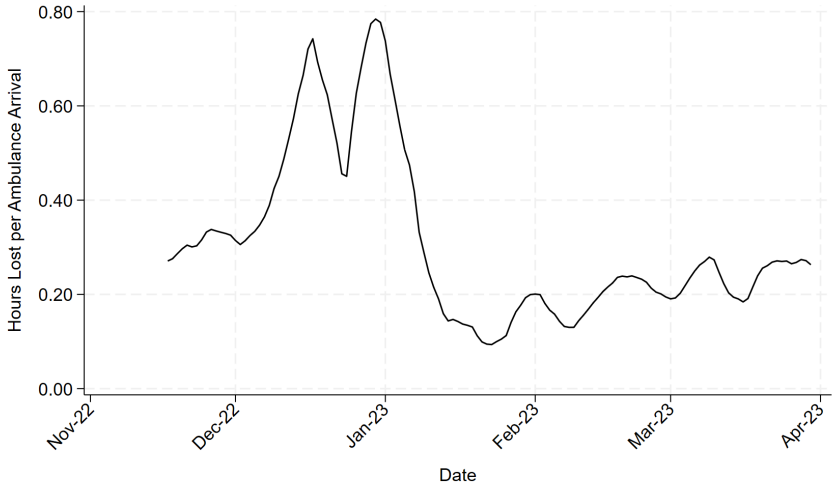


Figure B.23: Daily event study of total handover delay hours

